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# **Mortality Analysis Of Patients With Sepsis Associated Acute Kidney Injury (SAAKI) Cohort from MIMIC-IV dataset**

*A Thesis*

*Submitted by*

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# THESIS CERTIFICATE

This is to undertake that the Thesis titled **MORTALITY ANALYSIS OF PATIENTS WITH SEPSIS ASSOCIATED ACUTE KIDNEY INJURY (SAAKI) COHORT FROM MIMIC-IV DATASET**, submitted by me to the Indian Institute of Technology Madras, for the award of **Master Of Technology in Industrial Artificial Intelligence**, is a bona fide record of the research work done by me under the supervision of **Debopam Nanda & Samyabrata Chakraborty**. The contents of this Thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

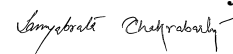
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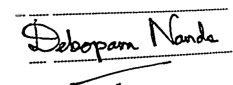
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# ABSTRACT

**KEYWORDS** sepsis-associated acute kidney injury (SA-AKI); sepsis-3; KDIGO; early warning; risk stratification; leakage-aware modeling; calibration; SHAP explanations; decision-curve analysis; survival analysis; MIMIC-IV; UMLS; RxNorm; SNOMED CT; ATC

This thesis addresses a critical, time-sensitive clinical question: at the end of the first ICU day (T0–T24), whether patients with sepsis-associated acute kidney injury (SA-AKI) who are at the highest risk of in-hospital death can be reliably identified, and whether the reasoning behind such predictions can be made transparent. Previous SA-AKI models, while often reporting strong discrimination, have been found to suffer from methodological issues that limit their clinical utility, including temporal leakage from the use of post-baseline data, global fitting of preprocessing steps, and a lack of model calibration. To overcome these limitations, a leakage-averse, TRIPOD-aligned framework was developed. An ontology-integrated data warehouse was built using PostgreSQL to harmonize the MIMIC-IV dataset with medical terminologies (UMLS, RxNorm, SNOMED CT, ATC). The SA-AKI cohort was rigorously defined using Sepsis-3 and KDIGO criteria, with all predictors strictly confined to the T0–T24 window. A one-time 80/20 stratified split ensured that all data-dependent steps, including a novel approach to creating and selecting significant missingness indicators and performing median imputation, were fitted only on training data. The methodology encompassed three stages: (i) an extensive exploratory data analysis on the training set to identify clinical phenotypes; (ii) supervised mortality prediction using a benchmark of models, from which a LightGBM classifier was selected, tuned with Optuna, and calibrated using an isotonic regressor on a validation set; and (iii) survival analysis to model time-to-death using the same early covariates with Cox Proportional Hazards and a custom DeepSurv model. On the held-out test set, the final calibrated LightGBM model achieved an AUROC of **0.749** and an F1-score of 0.522, indicating reliable discriminative ability.



The survival analysis yielded C-indexes of 0.69 (Cox) and 0.69 (DeepSurv). The primary contribution of this work is a reproducible, timing-faithful pipeline that delivers a calibrated, interpretable risk score and time-to-event estimates, positioning it as a credible tool for bedside clinical decision support.

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# CHAPTER 1

## INTRODUCTION

### 1.1 CLINICAL BACKGROUND AND MOTIVATION

Sepsis-associated acute kidney injury (SA-AKI) has emerged as one of the most consequential syndromes in critical care because it couples the systemic instability of sepsis with acute loss of renal function and thereby compounds risk across organ systems. For clinical and research purposes SA-AKI is commonly labeled when acute kidney injury develops within seven days of sepsis onset so that kidney dysfunction can be attributed to the septic process rather than to unrelated peri-ICU events (Peerapornratana *et al.*, 2019). The constituent conditions are defined by international consensus. Sepsis is defined by Sepsis-3 as life-threatening organ dysfunction due to a dysregulated host response to infection and is operationalized by an acute rise in SOFA score of two points or more (Singer *et al.*, 2016). Acute kidney injury is defined and staged by KDIGO using serum creatinine and urine output thresholds, including an absolute creatinine increase of at least 0.3 mg/dL within forty-eight hours, a relative increase of at least 1.5× baseline within seven days, or oliguria at or below 0.5 mL/kg/h for six hours (Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012). Within intensive care units SA-AKI has been reported to account for a large share of AKI cases and to be associated with sharply higher in-hospital mortality, longer length of stay, and increased risk of subsequent chronic kidney disease in survivors (Poston and Koyner, 2019; Peerapornratana *et al.*, 2019). These epidemiological features have kept early risk stratification at the forefront of research because timely identification of clinical deterioration is central to appropriate escalation, nephroprotective strategies, and resource planning.

## 1.2 WHAT THE LITERATURE HAS TRIED AND WHERE IT FALLS SHORT

Risk assessment in the ICU has traditionally relied on general severity scores such as SOFA, SAPS II, and APACHE II. These scores have offered useful global snapshots of illness severity but they tend to show only fair discrimination when the target is SA–AKI mortality, and they do not represent the nonlinear and interacting physiology that links infection biology, hemodynamics, renal injury, and treatment exposure in this population (Luo *et al.*, 2022; Nong *et al.*, 2024). In recent years the availability of large electronic health record repositories, notably MIMIC–IV, has encouraged the development of machine learning models that can take advantage of richer feature spaces and flexible function classes (Johnson *et al.*, 2023). Tree-based ensembles such as extreme gradient boosting frequently outperform baseline scores in internal testing and AUROC values near or above 0.80 are often reported (Chen *et al.*, 2025; Nong *et al.*, 2024; Wang *et al.*, 2024). Nevertheless, several methodological patterns recur across studies and create uncertainty about the true clinical utility of published models. Temporal anchoring of predictors is sometimes vague and whole-stay aggregation has been used, which allows future information to flow into features that are later presented as early predictors. Proxy variables for time to event, including length of stay, have been included as inputs to mortality models, which introduces a direct leakage channel from outcome process to covariates and inflates apparent discrimination (Chen *et al.*, 2025). Data-dependent steps such as feature selection and imputation have sometimes been performed across the full dataset rather than being learned on development folds and then applied to held-out data, which again invites optimistic estimates (Kang and Yoon, 2023). Class imbalance has often been handled through synthetic oversampling, including SMOTE and ADASYN, without careful assessment of whether the generated examples concentrate in overlapped or noisy regions or near outliers where decision boundaries are unstable (Chawla *et al.*, 2002; He *et al.*, 2008; Blagus and Lusa, 2013; Sáez *et al.*, 2015; Krawczyk, 2016). Calibration and principled threshold selection have been inconsistently reported, which limits bedside interpretability even when AUROC values appear strong. Finally, most



model development has relied on data from a single tertiary-care center, and performance losses under transport to other hospitals and populations have been documented, which tempers claims of broad generalizability (Johnson *et al.*, 2023; Rockenschaub *et al.*, 2024).

### **1.3 PROBLEM FOCUS AND RESEARCH OBJECTIVES**

The central challenge addressed in this thesis is the chasm between the promising internal accuracy reported in machine learning literature and the practical bedside utility of predictive models for SA–AKI mortality. This gap is primarily shaped by two issues: first, a failure to align predictive pipelines with the real-world clinical decision timeline, and second, a lack of transparency and calibration that hinders a model’s adoption. The objective of this work is to develop and validate a mortality risk stratification framework for SA–AKI that is methodologically sound, clinically relevant, and transparent. The work is conducted using the MIMIC–IV dataset (Johnson *et al.*, 2023), with a strict focus on an early clinical horizon to ensure the model’s predictions are actionable. The specific aims are:

1. To perform a systematic exploratory data analysis to identify key clinical phenotypes and risk factors within the training cohort.
2. To develop and validate a well-calibrated LightGBM classification model for in-hospital mortality using only features available within the first 24 hours of ICU admission.
3. To ensure model transparency by providing both global and local feature-level explanations using SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017).
4. To complement the binary prediction with a dynamic risk assessment by developing survival models using Cox Proportional Hazards and DeepSurv to estimate the *time to in-hospital death*.

## 1.4 PROPOSED APPROACH AND EVALUATION PLAN

The approach adopted in this thesis begins with the disciplined construction of an SA-AKI cohort according to Sepsis-3 and KDIGO criteria, with all predictors strictly limited to the first 24 hours (T0-T24) of an ICU stay (Singer *et al.*, 2016; Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012). This work is distinguished by a leakage-aware preprocessing pipeline where all steps were fitted exclusively on training data. This includes a formal test for the missing data mechanism, followed by the creation and statistical selection of informative missingness indicators and median imputation. A broad benchmark of machine learning models was conducted, leading to the selection of a final LightGBM model based on its superior performance. This model was tuned using the Optuna framework, explicitly calibrated using isotonic regression, and an optimal decision threshold was determined via Youden’s J statistic on a validation set. Explanations for the final model’s predictions were generated using SHAP to ensure transparency. Finally, to provide a dynamic view of risk over time, two survival models, a traditional Cox Proportional Hazards model and a neural network-based DeepSurv model, were developed and evaluated using the same early-window covariates. The evaluation plan prioritizes methodological transparency and clinical relevance, reporting a suite of metrics including AUROC, F1-score, Harrell’s C-index, and calibration plots on a strictly held-out test set, thereby addressing the common pitfalls identified in the literature (Kang and Yoon, 2023; Chen *et al.*, 2025).

## 1.5 CHAPTER ROADMAP

The remainder of the thesis is organized to follow this arc from clinical framing to deployment-oriented evaluation. Chapter 2 reviews the background in SA-AKI epidemiology and definitions, surveys conventional scores and machine learning approaches, and synthesizes methodological lessons from recent studies. Chapter 3 states the formal problem and gives the mathematical formulation for the binary prediction and survival analysis tasks. Chapter 4 provides a detailed description of the

cohort construction, the leakage-aware preprocessing pipeline, the model development and calibration process, and the evaluation protocol. Chapter 5 presents the findings from the extensive exploratory data analysis of the training cohort, identifying key clinical phenotypes and variable distributions that inform the modeling process. Chapter 6 presents the final performance results for the classification and survival models on the held-out test set, including discrimination, calibration, and SHAP-based interpretations. Finally, Chapter 7 concludes with a summary of findings, comments on clinical implications, and outlines future research directions.

## CHAPTER 2

### LITERATURE REVIEW

#### 2.1 CLINICAL DEFINITIONS AND EPIDEMIOLOGICAL CONTEXT

Sepsis-associated acute kidney injury (SA-AKI) has been characterized as the concurrence of sepsis and acute kidney injury with persistently poor outcomes. Sepsis has been defined by the Sepsis-3 consensus as life-threatening organ dysfunction arising from a dysregulated host response to infection, and the construct has been operationalized as suspected or documented infection together with an acute rise in organ failure, commonly a SOFA increase of at least two points (Singer *et al.*, 2016). Acute kidney injury has been defined by KDIGO through serum creatinine and urine output thresholds, in particular an absolute creatinine increase of at least 0.3 mg/dL within forty-eight hours, or at least 1.5× baseline within seven days, or urine output of 0.5 mL/kg/h or less for six hours (Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012). Many studies have labeled SA-AKI when AKI develops within a short interval after sepsis onset, frequently within seven days, to attribute renal dysfunction to the septic process rather than to competing causes (Peerapornratana *et al.*, 2019; Poston and Koyner, 2019). The shift toward Sepsis-3 and KDIGO has reduced heterogeneity relative to older criteria and has improved comparability across cohorts (Singer *et al.*, 2016; Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012).

The burden of SA-AKI in intensive care units is substantial. AKI develops in roughly fifty to sixty percent of septic ICU patients, and sepsis accounts for about forty to seventy percent of AKI in that setting (Peerapornratana *et al.*, 2019; Poston and Koyner, 2019). When AKI complicates sepsis, marked increases in in-hospital mortality follow, with earlier analyses suggesting a six to eight fold rise compared with sepsis without AKI and

mortality rates commonly between forty and sixty percent (Poston and Koyner, 2019; Peerapornratana *et al.*, 2019). Prolonged ICU stays and an elevated risk of subsequent chronic kidney disease have also been recorded among survivors (Poston and Koyner, 2019; Peerapornratana *et al.*, 2019). These epidemiological patterns have motivated the development of early risk stratification tools.

## **2.2 CONVENTIONAL SEVERITY INDICES AND EARLY WARNING SYSTEMS**

General ICU indices such as SOFA, SAPS II, and APACHE II have been evaluated for SA-AKI outcomes and tend to show only modest discrimination for short-term mortality or kidney-specific endpoints in this subgroup (Luo *et al.*, 2022; Li *et al.*, 2023; Nong *et al.*, 2024). Classical regression frameworks, including logistic regression and Cox models, have been employed repeatedly, yet difficulties persist in capturing the nonlinear and interacting relationships among infectious physiology, hemodynamics, renal injury, and treatment exposures. Performance has often failed to transport reliably across hospitals as a result (Kang and Yoon, 2023; Luo *et al.*, 2022; Li *et al.*, 2023). Earlier variability in sepsis and AKI definitions compounded the challenge by complicating cohort selection and cross-study comparison prior to widespread adoption of Sepsis-3 and KDIGO (Singer *et al.*, 2016; Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012).

## **2.3 DATASETS AND COHORT CONSTRUCTION IN SA-AKI MODELING**

Large electronic health record repositories have supported the development of prognostic models for SA-AKI, with MIMIC-IV serving as the most common source for cohort construction and feature derivation (Johnson *et al.*, 2023). Studies typically restrict analysis to the first ICU admission per patient, exclude patients younger than eighteen years, and remove ICU stays shorter than forty-eight hours. A central design decision involves the temporal window for feature construction. Fixed early windows, for example

the first twenty-four or forty-eight hours after admission or sepsis recognition, support timeliness, whereas whole-stay aggregation from admission to discharge has also appeared frequently and creates a risk that future information relative to an intended decision point enters the model inputs.

## **2.4 FEATURE ENGINEERING, SELECTION, AND LEAKAGE RISKS**

The handling of high-dimensional clinical variables has relied on screening by correlation or variance inflation factor, wrapper methods such as recursive feature elimination, and embedded strategies that exploit model-based importance estimates. It is common to find feature selection steps that were informed by the full dataset. When selection proceeds in this manner, the final predictor set is influenced by outcomes from records that are later evaluated, and optimism in reported discrimination is encouraged (Kang and Yoon, 2023). Claims that VIF-based pruning enhances stability and generalizability have also appeared without pre–post diagnostics or external tests, and the relevance of VIF is limited for classes of models that tolerate correlated predictors. In several studies, length of stay was included among features for mortality prediction. Because time to event is strongly associated with whether death occurred during the admission, that inclusion acts as a conduit for post-baseline information and dominates other covariates during training. Redundancy among variables has also been noted. For example, total urine output and average urine output become scalar multiples under a fixed observation horizon, and diverging importance profiles for those variables suggest that non-fixed or whole-stay windows were used, which reduces alignment with early prediction. Dependencies between creatinine and derived eGFR similarly complicate interpretation when both are included without a clear rationale.

## **2.5 MISSING DATA MECHANISMS AND IMPUTATION CHOICES**

The mechanisms that govern missingness have rarely been examined in depth in this literature. Assumptions that data were missing completely at random or missing at

random are often stated without empirical assessment, and simple percentage-based rules trigger imputation. Mean imputation for continuous variables remains a common choice even for biomarkers that show marked skew, such as serum creatinine, which is right skewed and well approximated by a log-normal distribution (Haeckel and Wosniok, 2010). Multiple imputation by chained equations and random forest variants is widely used, and in many cases those imputation models are fitted across the entire dataset prior to train–test separation. Because chained equations use all observed variables to impute each target variable, global fitting allows information from records later treated as held out to influence imputed values, and optimistic estimates of model performance follow.

## **2.6 CLASS IMBALANCE, EVALUATION STRATEGY, AND THRESHOLDING**

Mortality among SA–AKI cohorts is imbalanced, and class imbalance is most often handled by oversampling techniques, including SMOTE and ADASYN (Chawla *et al.*, 2002; He *et al.*, 2008). The two methods have occasionally been applied together even though ADASYN is itself a variant that emphasizes synthesis in hard regions of the feature space. SMOTE may generate synthetic points in overlapped or noisy areas, and performance can degrade in high-dimensional settings unless additional filtering is applied (Blagus and Lusa, 2013; Sáez *et al.*, 2015). Concentration of synthesis near low-density or borderline regions by ADASYN can shift decision boundaries around outliers and may harm stability under severe imbalance or distribution shift (Krawczyk, 2016). Comparisons with alternative imbalance strategies, for instance through intrinsic class weighting or hybrid sampling, have not always been provided. Evaluation strategies have often relied on a single split of the data into training and test sets without a separate validation split, and explicit calibration analyses and principled threshold selection have been omitted in several reports.

## **2.7 MODEL CLASSES, TUNING, CALIBRATION, AND INTERPRETABILITY**

Tree-based ensembles, notably extreme gradient boosting, have been adopted widely in this domain to represent nonlinearities and interactions among clinical variables, and stronger internal discrimination than baseline ICU scores is often reported (Chen and Guestrin, 2016; Luo *et al.*, 2022; Li *et al.*, 2023; Nong *et al.*, 2024; Wang *et al.*, 2024; Chen *et al.*, 2025; Jiang *et al.*, 2025). Hyperparameter tuning is usually performed through grid search cross validation, and claims of optimality appear despite the presence of complex, nonconvex search spaces in which only local minima are guaranteed by such procedures. Calibration is reported inconsistently, and explanations based on Shapley values are used to describe feature contributions, although stability of such explanations is not always explored. In several comparisons, confidence intervals for ensembles and penalized linear models overlap, and differences attributed to model class are entangled with choices about temporal windows, selection scope, and imputation design.

## **2.8 EXTERNAL VALIDITY, TRANSPORTABILITY, AND DEPLOYMENT TIMING**

Generalization beyond the MIMIC–IV development site remains a consistent concern. MIMIC–IV reflects a single tertiary-care center in Boston with local care patterns and population structure that are not broadly representative (Johnson *et al.*, 2023). External evaluations of ICU prediction models show loss of accuracy when transported across hospitals, pointing to center-specific differences that matter for performance (Rockenschaub *et al.*, 2024). Assertions about applicability to diverse populations therefore require cautious interpretation when external testing is absent. In addition, the timing of features relative to the intended decision point is often unspecified, and whole-stay aggregation creates ambiguity about whether a model could power actionable alerts at twenty-four or forty-eight hours.



## 2.9 DETAILED APPRAISAL OF REPRESENTATIVE STUDIES

### 2.9.1 Chen et al. (2025): SA-AKI mortality in MIMIC-IV

A recent contribution by Chen et al. reported an AUROC of 0.878 for mortality prediction in a SA-AKI cohort drawn from MIMIC-IV, and the result was presented as an improvement over the AUROC 0.794 attributed to Li et al. (Chen *et al.*, 2025; Li *et al.*, 2023). The cohort retained only the first ICU admission per patient and excluded patients younger than eighteen years and ICU stays shorter than forty-eight hours. The modeling pipeline combined VIF-based filtering with recursive feature elimination to produce a set of twenty-four predictors, after which several variables were reintroduced as expert inputs without quantitative justification. Because selection was described as having been performed on the entire dataset, the final predictor set was influenced by outcomes from records later treated as held out and the risk of information leakage was created (Kang and Yoon, 2023). A claim of generalizability across diverse populations was made, yet MIMIC-IV represents a single center and patient mix, and performance drift across hospitals has been documented (Johnson *et al.*, 2023; Rockenschaub *et al.*, 2024). The introduction criticized Li et al. for relying on forty-four variables on the grounds that the set might be too narrow to capture the complexity of SA-AKI, although the final model by Chen et al. used an even smaller set of twenty-four features, which created an internal inconsistency in the narrative (Chen *et al.*, 2025; Li *et al.*, 2023). Hyperparameters were tuned with grid search cross validation and described as optimal, although such procedures identify settings within a discrete grid and do not ensure global optimality in nonconvex spaces. Claims of timely, accurate, and actionable interventions were advanced, yet the temporal window for aggregating time-varying biomarkers was not specified. Because Sepsis-3 and KDIGO anchor observation to early horizons, an early extraction window would have been expected for actionable deployment; multiple signals suggested that whole-stay aggregation may have been used, which would introduce future information relative to early prediction. Missing values were imputed by means that were tied to percentage thresholds, and mean imputation was applied for variables with up to

twenty percent missingness. This approach is not well aligned with skewed distributions such as serum creatinine, which has been described as right skewed and is well modeled by a log-normal distribution (Haeckel and Wosniok, 2010). Missingness indicators were not created and the mechanism of missing data was assumed rather than studied. VIF filtering was described as enhancing stability and generalizability without pre–post diagnostics or external assessment and has limited relevance for ensembles that tolerate correlated predictors. The base estimator used within recursive feature elimination was not stated, and a linear estimator would impose a separability assumption during selection. Length of stay was included among predictors, and this inclusion introduced a strong proxy for time to event that is not available prospectively for early prediction. The simultaneous inclusion of total and average urine output suggested redundancy if a fixed window had been used, and differences in importance between the two variables implied that a non-fixed or whole-stay window was applied. The evaluation strategy used a seventy-five to twenty-five train–test split without a separate validation split, and explicit calibration and threshold selection were not described. Class imbalance was handled by applying SMOTE and ADASYN together, a combination that lacks a clear rationale and that may synthesize examples in overlapped or noisy regions or near outliers and thereby distort decision boundaries (Chawla *et al.*, 2002; He *et al.*, 2008; Blagus and Lusa, 2013; Sáez *et al.*, 2015; Krawczyk, 2016). Univariate comparisons between survivors and a group termed non-survivors were presented with t tests even though laboratory variables such as creatinine generally deviate from normality (Haeckel and Wosniok, 2010). The reported confidence intervals for ensemble and linear models overlapped, with AUROC 0.878 (95% CI 0.859 to 0.897) for extreme gradient boosting and AUROC 0.849 (95% CI 0.824 to 0.873) for logistic regression, which supported the interpretation that any advantage might be smaller than suggested once temporal alignment and leakage are addressed (Chen *et al.*, 2025).

### **2.9.2 Li et al. (2023): SA-AKI mortality in MIMIC-IV**

Li et al. assembled a SA-AKI cohort from MIMIC-IV by retaining only the first ICU admission and excluding ICU stays shorter than forty-eight hours. Features were extracted from the first twenty-four hours and were aggregated using maximum values, which provided a clear early window for model inputs (Li *et al.*, 2023). Lasso-regularized logistic regression was used for feature selection, and selection was described across the full dataset rather than within development folds. Because selection then involved outcomes from records later used for evaluation, optimism in reported performance was encouraged. The linear framework also prioritized main effects and could miss interactions and subtle nonlinear patterns that tree ensembles tend to capture. Multiple imputation by chained equations with random forest implementations was applied globally across the dataset rather than learned on development data and then applied to evaluation data, and missingness indicators were not constructed. The data were split into training and validation in an eighty to twenty ratio, and a discrimination gap favoring extreme gradient boosting over logistic regression was reported with AUROC 0.794 versus 0.730, although the magnitude of this gap remained difficult to interpret in light of the selection and imputation scope (Li *et al.*, 2023).

### **2.9.3 Hu et al. (2021): survival analysis in MIMIC-III**

A retrospective analysis by Hu et al. used MIMIC-III with smaller sample size and older coding structure to study SA-AKI outcomes (Hu *et al.*, 2021). The cohort retained only the first ICU admission and excluded patients younger than eighteen years and ICU stays shorter than forty-eight hours, and rows with more than five percent missing variables were removed. Lasso regression and Cox regression were used, and a C index of 0.72 was reported on a validation set. Biomarkers were aggregated by medians across the entire admission, therefore features included information that would not be available at an early decision point. Features with more than twenty percent missingness were dropped without investigation of mechanisms, and multiple imputation by chained equations was performed on the full dataset prior to splitting, which created leakage. The test

set contained one hundred and two rows, and reliability of performance estimates was limited by that size. Ethnicity was used as a predictor, which raised concerns about representativeness and fairness for transportability. Survival models at seven, fourteen, and twenty-eight days yielded AUCs of 0.77, 0.73, and 0.71 respectively.

#### **2.9.4 Roknaldin et al. (2024): sepsis to AKI prediction in MIMIC–III**

A recent preprint constructed models that predict AKI among septic ICU patients using MIMIC–III (Roknaldin *et al.*, 2024). Adults aged eighteen to eighty-nine years with sepsis by ICD–9 codes were included after excluding multiple admissions, ICU stays shorter than forty-eight hours, and records with more than twenty percent missingness. Features from the first twenty-four hours covered demographics, comorbidities, vitals, laboratory results, and early interventions. Multiple imputation was used for missing values, a correlation filter retained twenty-three predictors, and class imbalance was addressed with SMOTE. Six algorithms were compared, and logistic regression achieved the strongest internal results with test AUC 0.887 and validation AUC 0.857 together with a Brier score near 0.13 and acceptable isotonic calibration. Top predictors included urine output, bilirubin measures, blood urea nitrogen, minimum eGFR, weight, and indicators of vasopressor use and mechanical ventilation. The transparency of the feature list and the early window supported interpretability, while several design choices encouraged optimism. Feature selection and SMOTE were described prior to the final split into development and evaluation, so that the set of predictors and synthetic examples were influenced by records that later appeared in evaluation. Multiple imputation also appeared to have been fitted globally. A fixed correlation threshold risked removal of variables that are weakly correlated but jointly informative, and the inclusion of minimum and maximum eGFR alongside creatinine introduced redundancy because eGFR is derived from creatinine and demographics. For descriptive group comparisons, t tests were used for markers that are typically skewed, and the allocation of half the data to validation and one tenth to testing left a small test set. Calibration procedures were not tied to a recalibration step that excluded the test cohort, which limited the strength of calibration

claims on that set.

### **2.9.5 Chaudhary et al. (2020): phenotype discovery in MIMIC-III**

An unsupervised study in *CJASN* sought to identify SA-AKI phenotypes rather than predict mortality directly (Chaudhary *et al.*, 2020). An autoencoder was used to learn representations that were then clustered into three subgroups with distinct twenty-eight day mortality and dialysis rates. Differences were driven by bilirubin, lactate, transaminases, and inflammatory counts, and the highest risk subgroup showed markedly higher dialysis use and death. The analysis relied on whole-stay trajectories of laboratory and vital signs, which embeds information not available at early decision times. The sepsis cohort was constructed by codes, which can misalign the timing of the septic insult, and external validation of cluster stability was not performed. The exploratory links between clusters and treatments were not framed as causal and would require dedicated designs to support treatment guidance.

### **2.9.6 Lin et al. (2025): progression to chronic kidney disease in MIMIC-IV**

A study in *Informatics in Medicine Unlocked* examined progression from SA-AKI to incident chronic kidney disease using MIMIC-IV (Lin *et al.*, 2025). Extreme gradient boosting with SHAP explanations was trained on features collected during the first twenty-four hours, and high internal discrimination was reported. Thresholds for age, maximum creatinine, minimum hemoglobin, and lactate were described, and a feature stability analysis across AKI stages was presented. Although the early window and the inclusion of decision-curve analysis aided interpretability, variable screening by Lasso and stepwise logistic regression prior to final modeling, if performed across the full dataset, would have allowed outcomes from evaluation records to influence the screened set. Outcome assessment relied on follow up up to five years, and selection by loss to follow up as well as center-specific practice patterns could have influenced estimates. Thresholds based on SHAP dependence plots have known sensitivity to sampling variation, and replication would be expected for reliable adoption.

### **2.9.7 He et al. (2021): acute kidney disease after SA-AKI**

An article in *Frontiers in Medicine* predicted subsequent acute kidney disease rather than in-hospital mortality (He *et al.*, 2021). A single-center development cohort was used, and external validation was undertaken in MIMIC-III. Recurrent neural networks with long short-term memory units were compared with a decision tree and logistic regression. The recurrent model achieved an internal AUROC of 1.00 and strong performance in the external cohort, and the change in nonrenal SOFA between day one and day three emerged as an important signal. The development cohort contained two hundred and nine patients, which is a modest sample for a deep architecture with more than two dozen features, and perfect discrimination in such data is difficult to reconcile with typical ICU noise and suggests that overfitting, simplified labels, or other forms of information leakage may have been present. The dynamic window implied by the day one to day three delta improves clinical timing, while the time of alert generation relative to clinical workflow was not specified. It was not always clear that imputation and preprocessing steps were confined to development data.

### **2.9.8 Kantola et al. (2025): admission procalcitonin and risk of AKI**

A multicenter retrospective cohort across three hospitals evaluated admission procalcitonin for prediction of subsequent AKI and reported an odds ratio of approximately 1.01 per ng/mL with poor overall discrimination near ROC AUC 0.59 (Kantola *et al.*, 2025). The heterogeneity of practical cutoffs across prior work and the inconsistent relationships between procalcitonin levels and AKI severity were emphasized, and the conclusion favored limited standalone clinical utility for early risk stratification. Procalcitonin is not available in MIMIC-IV, which restricts replication in that database.

## 2.10 SYNTHESIS AND IMPLICATIONS FOR THIS THESIS

The literature reviewed above sketches a consistent picture in which window definition, the scope of data-dependent preprocessing, handling of missingness and imbalance, and the availability of external evidence play decisive roles in the credibility of reported scores. Studies that relied on whole-stay aggregation, on inclusion of length of stay or other post-baseline correlates, and on global feature selection or imputation tend to report stronger discrimination, yet those gains are paired with ambiguity about practical timing and with higher risk of optimism. When early windows are enforced clearly, performance remains competitive and the path to bedside use becomes clearer. Several reports present overlapping confidence intervals between ensembles and penalized linear models, which suggests that differences in pipeline design often overshadow differences in the learning algorithms themselves (Kang and Yoon, 2023). Generalization beyond a single tertiary-care center remains a key uncertainty, and drift across hospitals is documented (Johnson *et al.*, 2023; Rockenschaub *et al.*, 2024). In this thesis, these observations are treated as design constraints for methodological choices, and the subsequent chapters are organized to respond to the gaps identified here while keeping the emphasis on temporally anchored modeling, leakage-aware preprocessing, calibrated evaluation, and transparency about what information is and is not available at the moment when a clinical decision is intended to be supported.

## CHAPTER 3

### PROBLEM DEFINITION AND FORMULATION

#### 3.1 CLINICAL DECISION AND SCOPE

Sepsis-associated acute kidney injury (SA-AKI) constitutes a high-risk state coupling infection, shock physiology, and acute renal dysfunction. The *decision point* is fixed: at the end of the first ICU day ( $T_{\text{obs}}=24$  h after ICU admission), the risk of in-hospital death is to be estimated for patients meeting SA-AKI criteria using only information available by  $T_{\text{obs}}$ . The primary goal was to develop a predictive model that is both *discriminative* and *calibrated*. Two auxiliary objectives support this decision: (i) systematic *exploratory data analysis* to discover clinical phenotypes in the early feature space, and (ii) *time-to-event* modeling that translates the same early covariates into survival functions for *time to death*. All cohort rules adhere to Sepsis-3 for sepsis and KDIGO for AKI, and reporting follows the TRIPOD statement.

**Out of scope.** This study does not estimate causal treatment effects or recommend specific therapies. The problem is predictive; models summarize risk conditional on the information set  $\mathcal{I}(T_{\text{obs}})$ .

#### 3.2 OBSERVATIONAL UNIT, CLOCKS, AND INFORMATION SETS

Each observational unit corresponds to the *first* eligible ICU admission for a patient in the SA-AKI cohort. Let ICU admission be  $t=0$  and define the early feature window as

$$[0, T_{\text{obs}}], \quad T_{\text{obs}} = 24 \text{ hours.}$$

All time-varying signals in this window—vital signs, laboratory measurements, fluids/urine output, and early therapies—are summarized by deterministic, leakage-safe



functionals into a feature vector

$$x_i \in \mathbb{R}^p,$$

augmented with static demographics and comorbidity indicators. For the binary classification task, a one-time stratified 80/20 train-test split yielded training and test sets with (8028, 162) and (2008, 162) dimensions, respectively (including the target column). The positive (death) rate in the training set was 26.74%. The survival analysis used the same patient split with a feature matrix of (8028, 124) and (2008, 124) before the inclusion of time and event columns.<sup>1</sup> The available information for patient  $i$  at the decision time is the  $\sigma$ -field

$$\mathcal{I}_i(T_{\text{obs}}) = \{\text{EHR observations with time stamp} \leq T_{\text{obs}}\}.$$

*Temporal discipline.* Any statistic requiring  $t > T_{\text{obs}}$  (e.g., total ICU length of stay, whole-stay extrema/means, post-baseline therapies) is deemed inadmissible for prediction and is excluded by a scripted leakage audit.

**Target population and case definition.** Sepsis is operationalized by Sepsis-3 (suspected infection plus a daily SOFA score increase of  $\geq 2$ ); AKI is defined by KDIGO serum-creatinine and urine-output criteria. SA-AKI requires AKI onset within  $\pm 48$  h of sepsis onset. Only adults ( $\geq 18$  y) with an ICU length of stay  $\geq 24$  h, and only the first ICU stay per patient, are included. Cohort definition timestamps  $t_{\text{sepsis}}$  and  $t_{\text{aki}}$  serve only to identify the cohort and *are not used as predictors*.

### 3.3 DATA-GENERATING VIEW AND NOTATION

Let  $(X, Y, T^{\text{death}}, \delta^{\text{death}}) \sim P$  on

$$\mathcal{X} \times \{0, 1\} \times \mathbb{R}_+ \times \{0, 1\},$$

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<sup>1</sup>These dataset sizes and event fractions reflect the final data preparation used in model building and survival analysis, as detailed in the notebooks.

where  $X$  is the T0–T24 feature vector;  $Y$  is in–hospital death (1/0);  $T^{\text{death}}$  is time from ICU admission to in–hospital death or censoring at discharge; and  $\delta^{\text{death}}$  is the event indicator (1=death, 0=censored). Independent, identically distributed samples  $\{(x_i, \cdot)\}_{i=1}^n$  are observed. A one–time 80/20 split yields  $(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}})$ . Any internal validation split for tasks like hyperparameter tuning or calibration is carved out exclusively *from within*  $\mathcal{D}_{\text{train}}$ . The survival analysis reuses the same patient split with right censoring.

### 3.4 PROBLEM 1 (UNSUPERVISED): EXPLORATORY ANALYSIS AND PHENOTYPING

**Goal.** To systematically explore the early feature space of the training data to identify clinically relevant patient phenotypes without using outcome labels. This involves analyzing the distributions of key biomarkers, assessing correlations between features and the mortality outcome, and comparing clinical characteristics (e.g., vital signs, lab values, SOFA components) between patients who survived and those who did not. This exploratory phase informs feature importance and provides clinical context for the supervised models.

### 3.5 PROBLEM 2 (SUPERVISED, PRIMARY): EARLY–WINDOW MORTALITY RISK

#### 3.5.1 Label and objective

Define the binary label

$$Y = \mathbb{I}\{\text{in–hospital death prior to discharge}\}.$$

A probabilistic classifier  $f_\theta : \mathcal{X} \rightarrow [0, 1]$  is sought such that

$$\hat{p}(x) = f_\theta(x) \approx \Pr(Y = 1 \mid X = x, \mathcal{I}(T_{\text{obs}})).$$

The learning criterion on training data is the average negative log-likelihood (cross-entropy),

$$\hat{\theta} \in \arg \min_{\theta} \frac{1}{m} \sum_{j=1}^m \{ -y_j \log f_{\theta}(x_j) - (1 - y_j) \log(1 - f_{\theta}(x_j)) \},$$

optimized using cross-validation within  $\mathcal{D}_{\text{train}}$ .

### 3.5.2 Performance targets

Two properties are co-primary:

- (a) **Discrimination:** High Area Under the Receiver Operating Characteristic Curve (AUROC) and F1-score on the held-out test set.
- (b) **Calibration:** Reliable probabilities, assessed by generating a final calibrated model and evaluating its performance.

Calibration is performed on a validation split drawn from  $\mathcal{D}_{\text{train}}$  using isotonic regression; final probabilities are  $\tilde{p} = c \circ f_{\theta}(x)$ , where  $c$  denotes the calibrator.

### 3.5.3 Decision rule and operating point

Clinical actions are thresholded via  $\delta_{\tau}(x) = \mathbb{I}\{\tilde{p}(x) \geq \tau\}$ . The optimal operating point  $\tau$  is selected on an internal validation split by maximizing Youden’s  $J$  index. The final model’s utility across a range of thresholds is also assessed using decision-curve analysis.

### 3.5.4 Hypothesis classes and regularization

A spectrum of models,  $\mathcal{H}$ , is benchmarked to assess performance across different functional capacities: Logistic Regression (with and without class weights, and with various resampling techniques), regularized models (Ridge, Lasso, ElasticNet), Linear and Quadratic Discriminant Analysis (LDA/QDA), Gaussian Naive Bayes, Decision Tree, Random Forest, gradient-boosted trees (XGBoost, LightGBM), and a deep neural network (DNN++). Hyperparameters for tree-based models are tuned using the Optuna framework within training folds.

### 3.5.5 Interpretability requirement

For the final selected model ( $f_\theta$ ), feature contributions are explained using SHapley Additive exPlanations (SHAP). This provides both global feature importance rankings and local, instance-level explanations for individual patient predictions, satisfying the need for model transparency.

**Constraints and guardrails.** (i) *Temporal*: All predictive features are derived strictly from the  $[0, 24]$  h window post-ICU admission. (ii) *Independence*: Only the first ICU stay per patient is included. (iii) *Siloed preprocessing*: All data transformations (imputation, scaling, indicator creation) are learned on the training set and applied without modification to the test set. (iv) *No peeking*: The test set is held out and used only for a single final evaluation. (v) *Cohort times not used as predictors*:  $t_{\text{sepsis}}$  and  $t_{\text{aki}}$  are used only for case definition.

## 3.6 PROBLEM 3 (SURVIVAL): EARLY-COVARIATE TIME TO DEATH

Using the same T0–T24 covariates, the *time to in-hospital death* is modeled with right censoring at hospital discharge. For the event time data  $(T^{\text{death}}, \delta^{\text{death}})$ , the conditional hazard function  $h(t \mid x)$  is estimated via:

- **Cox Proportional Hazards (PH)**: A semi-parametric model specified as  $h(t \mid x) = h_0(t) \exp\{\beta^\top x\}$ , implemented with the `lifelines` library. Proportional hazards assumptions are checked.
- **DeepSurv**: A neural network-based Cox model where  $h(t \mid x) = h_0(t) \exp\{g_\psi(x)\}$ , with  $g_\psi$  parameterized by a custom deep network implemented in PyTorch.

Performance is evaluated using Harrell’s  $C$ -index, time-dependent AUC, calibration plots at clinically relevant horizons (e.g., 90, 180, 365 hours), and Kaplan-Meier curve stratification by predicted risk.

### 3.7 EVALUATION PROTOCOL

For the classification task, a one-time 80/20 train–test split is used. An internal validation split is carved from the training set for model selection, calibration, and thresholding. The frozen pipeline is then applied once to the test set. For the survival task, the same patient partitions are used. Primary test metrics for the classifier include AUROC, F1-score, precision, and recall. For the survival models, primary metrics are Harrell’s *C*-index, calibration, and risk stratification confirmed by the log-rank test.

### 3.8 ASSUMPTIONS AND RISKS

- **Availability:** All predictors are assumed to be observable by  $T_{\text{obs}}$ ; features with excessive missingness or not typically available early are excluded.
- **Measurement:** Units are harmonized and physiological plausibility is enforced. The missing data mechanism is assumed to be Missing at Random (MAR) after finding it is not Missing Completely at Random (MCAR), justifying the use of imputation and missingness indicators.
- **Domain:** Models are developed on the MIMIC–IV dataset; transportability to external centers may require recalibration.
- **Predictive, not causal:** The models identify predictive associations, not causal relationships.

### 3.9 COMPACT STATEMENT OF AIMS

**Aim A (Phenotyping).** To conduct a thorough exploratory data analysis on the T0–T24 training data to identify and visualize the key clinical characteristics and phenotypes that differ between survivors and non-survivors.

**Aim B (Primary classifier).** To train and validate a calibrated probabilistic model for in–hospital death,  $\tilde{p}(x) = c \circ f_{\theta}(x)$ , ensuring high discrimination and utility through a strict, leakage-controlled pipeline.

**Aim C (Survival).** Using the same early covariates, to estimate the survival function

$S^{\text{death}}(t \mid x)$  with Cox and DeepSurv models and evaluate their performance via C-index, calibration, and risk stratification.

# CHAPTER 4

## METHODOLOGY

### 4.1 OVERVIEW

This chapter details the methodological pipeline that was designed to develop and validate predictive models for SA–AKI mortality. The workflow begins with the construction of a high-fidelity cohort from an ontology-integrated data warehouse and proceeds through three analytical stages: (i) exploratory data analysis for phenotyping, (ii) development of a supervised binary classifier for mortality (the primary task), and (iii) time-to-event survival analysis. A strict, one-time stratified 80/20 data split underpins the entire process to prevent information leakage. All data-dependent transformations were fitted on the training set only and subsequently applied to the test set for a single final evaluation. The development process was guided by the TRIPOD statement for transparent reporting.

### 4.2 DATA SOURCES, ETL, AND COHORT (T0–T24)

**ETL and Integration.** The analysis asset was built by ingesting and integrating multiple sources, primarily MIMIC–IV, with medical ontologies (RxNorm, ATC, SNOMED CT) linked via the Unified Medical Language System (UMLS). This was managed in a PostgreSQL database, enabling reproducible, ingredient-level value sets for concepts like systemic antibiotics and vasopressors. A validation suite ensured unit harmonization (e.g.,  $F \rightarrow C$ ,  $lb \rightarrow kg$ ), referential integrity, and logical consistency.

**Cohort Definition.** The study population was defined as adults ( $\geq 18$  y) on their first ICU stay with a length-of-stay  $\geq 24$  h. The SA–AKI cohort was identified using established clinical criteria:

- **Sepsis (Sepsis–3):** Defined by suspected infection (a body-fluid culture near the administration of a systemic antibiotic) and an acute increase in organ dysfunction

( $\Delta\text{SOFA} \geq 2$ ). The sepsis onset time,  $t_{\text{sepsis}}$ , was the start of the first qualifying ICU day.

- **AKI (KDIGO 2012):** Defined by changes in serum creatinine (SCr) or urine output. A hierarchical method established baseline SCr. The AKI onset time,  $t_{\text{aki}}$ , was the earliest time any criterion was met.
- **SA-AKI:** Required AKI onset within the window  $[t_{\text{sepsis}} - 48 \text{ h}, t_{\text{sepsis}} + 48 \text{ h}]$ .

Patients with prior End-Stage Renal Disease (ESRD) or who received Renal Replacement Therapy (RRT) within the first 24 h were excluded. All predictive features were restricted to measurements within the first 24 hours of ICU admission (T0–T24) to ensure clinical actionability and prevent look-ahead bias.

### 4.3 DATA SPLIT AND LEAKAGE CONTROL

A one-time, stratified 80/20 train-test split was performed on the cohort. Let  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$  be the full dataset, where  $x_i \in \mathbb{R}^P$  are the T0–T24 features and  $y_i \in \{0, 1\}$  is the in-hospital death indicator. The partition is

$$\mathcal{D} = \mathcal{D}_{\text{train}} \dot{\cup} \mathcal{D}_{\text{test}},$$

where partitions are disjoint and stratified by the outcome to maintain class balance. All data-dependent objects—imputation models, scaling parameters, and encoding vocabularies—were fitted *only* on  $\mathcal{D}_{\text{train}}$  and then applied as frozen transformers to  $\mathcal{D}_{\text{test}}$ . Any validation set used for model tuning or calibration was created as a further split *from within*  $\mathcal{D}_{\text{train}}$ .

**Leakage Audit.** A scripted audit ensured that no variables encoding information after T24 were included as predictors. This included removing features like total ICU length of stay and aggregates (e.g., `_n` observation counts) that could act as proxies for care intensity or illness duration.



#### 4.4 PREPROCESSING AND FEATURE ENGINEERING (TRAIN-ONLY)

**Missingness Analysis.** The nature of missing data was formally investigated on  $\mathcal{D}_{\text{train}}$ . Little’s MCAR test was performed, and its rejection of the null hypothesis suggested that data were not Missing Completely at Random (MCAR). This justified a more careful handling of missingness beyond simple imputation.

**Missingness Indicators.** For every feature with missing values, a binary indicator variable (`_missing`) was created. The association between each indicator and the mortality outcome was tested using a chi-square test. Only indicators that showed a statistically significant association ( $p < 0.05$ ) were retained and appended to the dataset as new features, thereby preserving potentially prognostic information contained in the pattern of missingness.

**Imputation.** After adding significant missingness indicators, remaining missing values in numerical columns were imputed using the median calculated from the training set.

**Feature Selection and Scaling.** Highly correlated features (Pearson correlation  $\geq 0.99$ ) were identified, and one from each pair was removed to reduce multicollinearity. All numerical features were then standardized using `StandardScaler`, with the scaler being fit only on the training data.

#### 4.5 EXPLORATORY DATA ANALYSIS AND PHENOTYPING (TRAINING ONLY)

An extensive EDA was performed on  $\mathcal{D}_{\text{train}}$  to characterize the cohort’s clinical properties. This served as an unsupervised phenotyping step. Key analyses included:

- **Distributional Analysis:** Visualizing the distributions of key continuous variables (e.g., age, APACHE III, SOFA score, creatinine, lactate) and comparing them between survivor and non-survivor groups using histograms and density plots.
- **Correlation Analysis:** Calculating the Pearson correlation of all features with the mortality outcome to identify the strongest univariate predictors.

- **Stratified Analysis:** Using violin plots, box plots, and heatmaps to compare vital signs, laboratory values, and SOFA sub-scores between outcomes. Statistical significance was assessed using the Mann-Whitney U test due to the non-normal distribution of most variables.

#### 4.6 SUPERVISED BINARY MORTALITY PREDICTION (PRIMARY)

**Learning Objective.** For a model  $f_\theta : \mathbb{R}^p \rightarrow [0, 1]$ , the goal was to minimize the average binary cross-entropy loss on  $\mathcal{D}_{\text{train}}$  using 5-fold cross-validation.

**Model Benchmarking.** A wide set of models was benchmarked: Logistic Regression, LDA, QDA, Naive Bayes, regularized linear models, Decision Tree, Random Forest, XGBoost, and LightGBM. For tree-based ensembles, hyperparameters were tuned using Optuna. The impact of class imbalance was addressed primarily through `class_weight='balanced'` and was compared against various oversampling strategies (e.g., SMOTE, ADASYN) in sensitivity analyses.

**Final Model Selection, Calibration, and Thresholding.** The LightGBM model was selected based on its superior performance. The final pipeline involved:

1. Training the optimized LightGBM model on a sub-split of the training data.
2. Calibrating the model's outputs on a held-out validation split (from the training set) using `CalibratedClassifierCV` with the isotonic method.
3. Determining the optimal classification threshold on the same validation split by maximizing Youden's J statistic.
4. Retraining the final, calibrated LightGBM model on the entire training dataset ( $\mathcal{D}_{\text{train}}$ ).
5. Evaluating the final model on the untouched test set ( $\mathcal{D}_{\text{test}}$ ) using the predetermined threshold.

**Interpretability.** The final LightGBM model was interpreted using SHAP. A beeswarm summary plot was generated to visualize global feature importance and the direction of effects.

## 4.7 SURVIVAL ANALYSIS (TIME TO DEATH)

The same preprocessed T0–T24 feature matrix was used to model time to in-hospital death, with patients discharged alive being right-censored.

- **Cox Proportional Hazards (PH):** A CoxPH model was fitted using the `lifelines` library. Feature selection was performed by retaining the top 50 predictors from an initial full model, and the penalizer was tuned. Results were interpreted via a forest plot of hazard ratios.
- **DeepSurv:** A custom neural network-based Cox model was implemented in PyTorch, using SELU activations and residual connections. The model was trained by minimizing the negative Cox partial log-likelihood with the Efron method for handling ties.

Model performance was assessed by Harrell’s C-index, time-dependent AUC, calibration plots at clinically relevant horizons (90, 180, and 365 hours post-admission), and Kaplan-Meier curves stratified by model-predicted risk quartiles. The statistical significance of the risk stratification was confirmed with the log-rank test.

## 4.8 ADHERENCE TO TRIPOD GUIDELINES

### 4.8.1 Data Leakage: Definition, Mechanisms, and Mitigation

In the context of clinical prediction modeling, data leakage has been defined as the inadvertent incorporation of information that would not be available at the intended time of prediction into the model training or evaluation process, resulting in inflated performance estimates that fail to reflect real-world utility (Kang and Yoon, 2023). For SA–AKI mortality prediction, three principal mechanisms through which leakage may be introduced have been identified in the literature and were considered during the design of this pipeline.

The first mechanism, temporal leakage (also termed look-ahead bias), arises when features are derived from measurements recorded after the intended prediction horizon. A model designed to estimate mortality risk at 24 hours post-admission that incorporates laboratory values from subsequent days, or whole-stay aggregates such as the mean

creatinine across the entire ICU episode, has been provided with information that encodes the patient’s future trajectory and outcome. The inclusion of total ICU length of stay as a predictor represents a particularly direct form of this bias, as length of stay serves as a strong proxy for whether and when death occurred (Chen *et al.*, 2025).

The second mechanism, preprocessing leakage, has been observed when data-dependent transformations (including imputation, scaling, and feature selection) are fitted on the entire dataset prior to the train–test split. Under such conditions, the statistical properties of the held-out records (means, variances, and correlations with the outcome) are permitted to influence the training pipeline, and optimistically biased performance estimates are produced as a consequence (Kang and Yoon, 2023).

The third mechanism, selection leakage, occurs when feature screening is conducted across the full dataset rather than within the training partition alone. Features that happen to correlate with the outcome across all records, including those subsequently reserved for evaluation, are preferentially retained, and the resulting predictor set carries information from the evaluation data.

In the present work, each of these mechanisms has been addressed through explicit safeguards. Temporal leakage has been prevented by restricting all features to the T0–T24 window and by deploying a scripted audit that flags any variable requiring post-T24 information. Preprocessing leakage has been avoided by fitting all transformers (imputer, scaler, and missingness indicator selector) exclusively on  $\mathcal{D}_{\text{train}}$ . Selection leakage has been precluded by confining all feature screening to the training partition. The held-out test set was used exactly once for final evaluation, and no iterative refinement based on test-set performance was undertaken.

## **4.8.2 The 24-Hour Window: Rationale, Strengths, and Heterogeneity**

### **Considerations**

The selection of a fixed 24-hour observation window anchored to ICU admission (T0–T24) was motivated by the structure of clinical workflows in intensive care. The first ICU day is the period during which the initial assessment is typically completed, baseline investigations become available, and early management decisions regarding fluid resuscitation, vasopressor initiation, and antibiotic escalation are made. A prediction generated at T24 is therefore aligned with the natural decision point at which escalation or de-escalation of care is considered.

A fixed clock anchored to ICU admission, however, introduces a source of heterogeneity that warrants explicit discussion. Patients arrive in the ICU at different stages of their illness trajectory: some are admitted directly from the emergency department near the onset of sepsis, whereas others are transferred from general wards after a period of progressive deterioration. Consequently, the T0–T24 window captures different phases of the underlying disease process across patients. For early presenters, the window may coincide with the initial inflammatory surge, while for late transfers, it may reflect a more established pattern of organ dysfunction.

This heterogeneity represents a genuine limitation of any admission-anchored design, yet it also reflects the conditions under which a bedside prediction tool would be required to operate in practice. A clinician querying the model at the 24-hour mark has access only to the information accumulated since admission, regardless of the patient's prior trajectory. Several features included in the model were observed to partially account for this variation: the APACHE III score, computed at admission, integrates acute physiology with chronic health status and thereby captures elements of the pre-ICU trajectory; the SOFA score at 24 hours reflects cumulative organ dysfunction irrespective of when it began; the Charlson comorbidity index components provide context regarding baseline vulnerability; and the first/last/delta/slope suffixes applied to time-varying

features encode within-window dynamics that have been expected to differ between early and late presenters.

Stratified analyses by admission source (emergency department versus ward transfer) or by the interval from symptom onset to ICU admission could be pursued in future work to quantify the degree to which model performance varies across these subgroups. An alternative design in which the observation window is anchored to sepsis onset rather than ICU admission has also been considered, though such an approach introduces its own difficulties because the sepsis onset time is itself an estimated quantity derived from clinical rules rather than a directly observed event.

The present study was designed in accordance with the TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) guidelines.

The key items addressed include:

- **Source of Data (Item 4a):** The study design and source of data (MIMIC-IV, a large electronic health record database) are clearly described.
- **Participants (Item 5a-b):** The study setting (ICU), location, and detailed eligibility criteria for participants are specified in the cohort construction.
- **Outcome and Predictors (Items 6a, 7a):** The outcome (in-hospital mortality) and all predictors (derived from the T0–T24 window) are explicitly defined.
- **Sample Size (Item 8):** The flow of participants through the study is described, detailing exclusions and the final sample size available for analysis.
- **Missing Data (Item 9):** The handling of missing data is detailed, including statistical testing of the missingness mechanism, creation and selection of significant missingness indicators, and median imputation based on the training set.
- **Statistical Analysis Methods (Item 10a-d):** Predictor handling (correlation removal, scaling), model-building procedures (benchmarking of multiple algorithms, Optuna for tuning), methods for internal validation (stratified 80/20 train-test split), and all performance measures (AUROC, F1-score, C-index, calibration plots, DCA) are specified.
- **Model Specification and Performance (Items 15a, 16):** The final model (LightGBM) and its key hyperparameters are presented. Performance measures on

the test set are reported, providing an unbiased estimate of the model's performance on new data.

## CHAPTER 5

### DATASET UNDERSTANDING AND EDA

#### 5.1 OVERVIEW

The data assets and clinical labeling rules that underlie the analysis-ready SA–AKI cohort are described first, followed by the temporal conventions and harmonization steps that precede the main analysis. The remainder of the chapter is devoted to the Exploratory Data Analysis (EDA) conducted on the training set. That EDA serves as a form of unsupervised phenotyping: it identifies the clinical characteristics and risk factors that differentiate patient outcomes and provides the empirical foundation for the predictive modeling that follows.

#### 5.2 SOURCE TABLES AND WAREHOUSE CONTEXT

Figure 5.1 positions the raw inputs within MIMIC–IV (core, hosp, ICU) that were ingested and optimized in the warehouse. These tables provide the vitals, labs, medications, microbiology, and encounter scaffolding used for labeling and features.

#### 5.3 ONTOLOGY-DRIVEN CONCEPT IDENTIFICATION

To avoid brittle, hard-coded drug lists, clinical concepts were mapped across terminologies and ingredient-level value sets were derived programmatically. The crosswalk in Figure 5.2 illustrates the pathway from SNOMED CT through UMLS CUIs into RxNorm ingredients, with ATC used as an orthogonal cross-reference and coverage check. This pipeline underlies the systemic *antibiotic* and *vasopressor* flags used in labeling and feature construction.



# The MIMIC-IV Data Universe

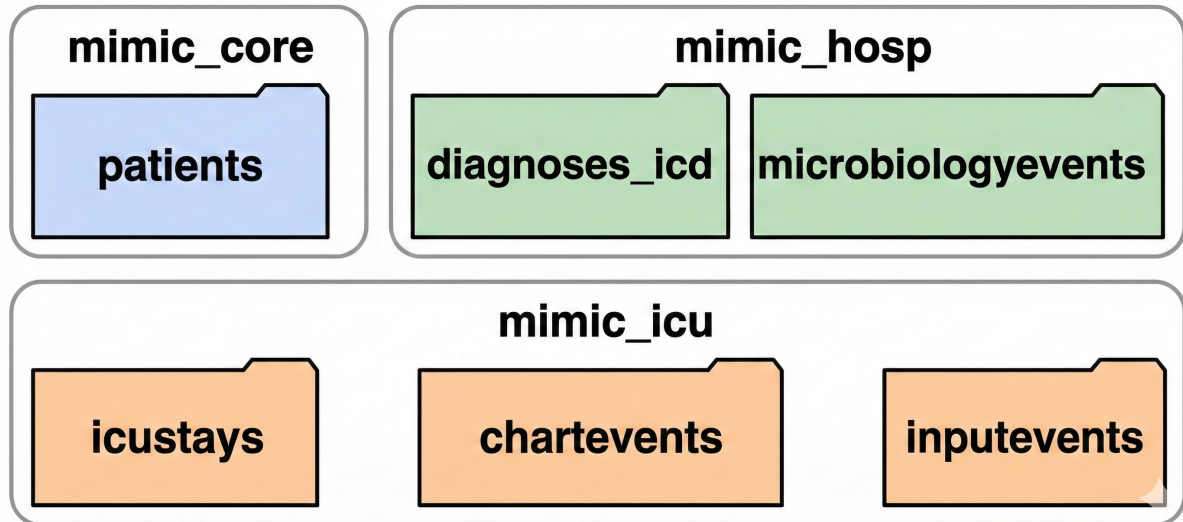


Figure 5.1: MIMIC-IV data universe leveraged in this study.

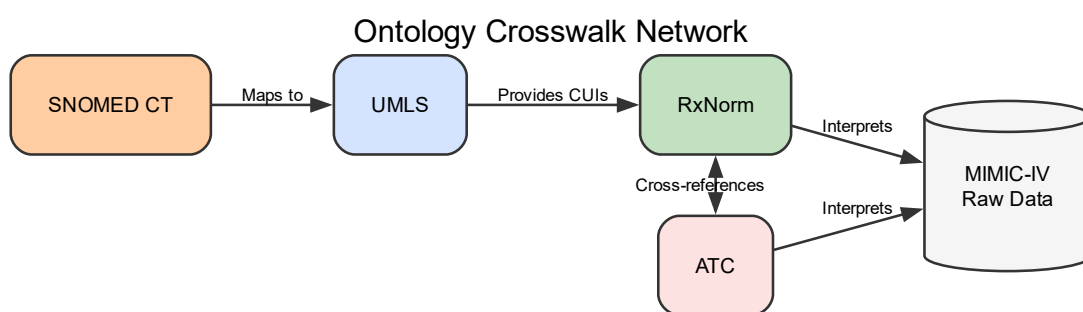


Figure 5.2: Ontology crosswalk linking SNOMED CT → UMLS CUIs → RxNorm ingredients, with ATC cross-reference.

## 5.4 COHORT CONSTRUCTION (COUNTS AND EXCLUSIONS)

A CONSORT-style reduction from all ICU stays to the final analysis cohort was followed. Figure 5.3 consolidates inclusion criteria (adult, first ICU stay, LOS  $\geq 24$  h), Sepsis-3 screening, SA-AKI temporal linkage, and final exclusions (ESRD, early RRT). These counts serve as the reference for all summary denominators used in the EDA.

## 5.5 CLINICAL LABELING RULES AND TIME ANCHORS

### 5.5.1 Sepsis-3 and AKI Logic (Compact View)

Sepsis-3 requires *suspected infection and* an acute increase in organ dysfunction ( $\Delta\text{SOFA} \geq 2$ ). AKI follows KDIGO using three criteria: serum creatinine (absolute/relative) and urine output (UO). The Sepsis-3 gate and the UO sessionization used for the KDIGO UO criterion are displayed as a two-panel composite in Figure 5.4.

### 5.5.2 Timeline Conventions and Windows

All predictors used in modeling are restricted to the first 24 h after ICU admission (T0-T24). Suspected infection is defined by a temporal link between cultures and systemic antibiotics, and SA-AKI requires AKI onset within  $\pm 48$  h of Sepsis onset. Figure 5.5 illustrates these anchors and windows.

## 5.6 DATA HARMONIZATION AND INITIAL STATISTICS

Before any analysis, data were harmonized (e.g., Fahrenheit  $\rightarrow$  Celsius), redundant features were removed, and comorbidities were consolidated. The final analytic cohort contained **n=10,036** ICU stays. A stratified 80/20 split resulted in a training set of 8,028 patients and a test set of 2,008 patients. The in-hospital mortality rate in the training set was 26.74%. Key baseline statistics are presented in Table 5.1 and Table 5.2.

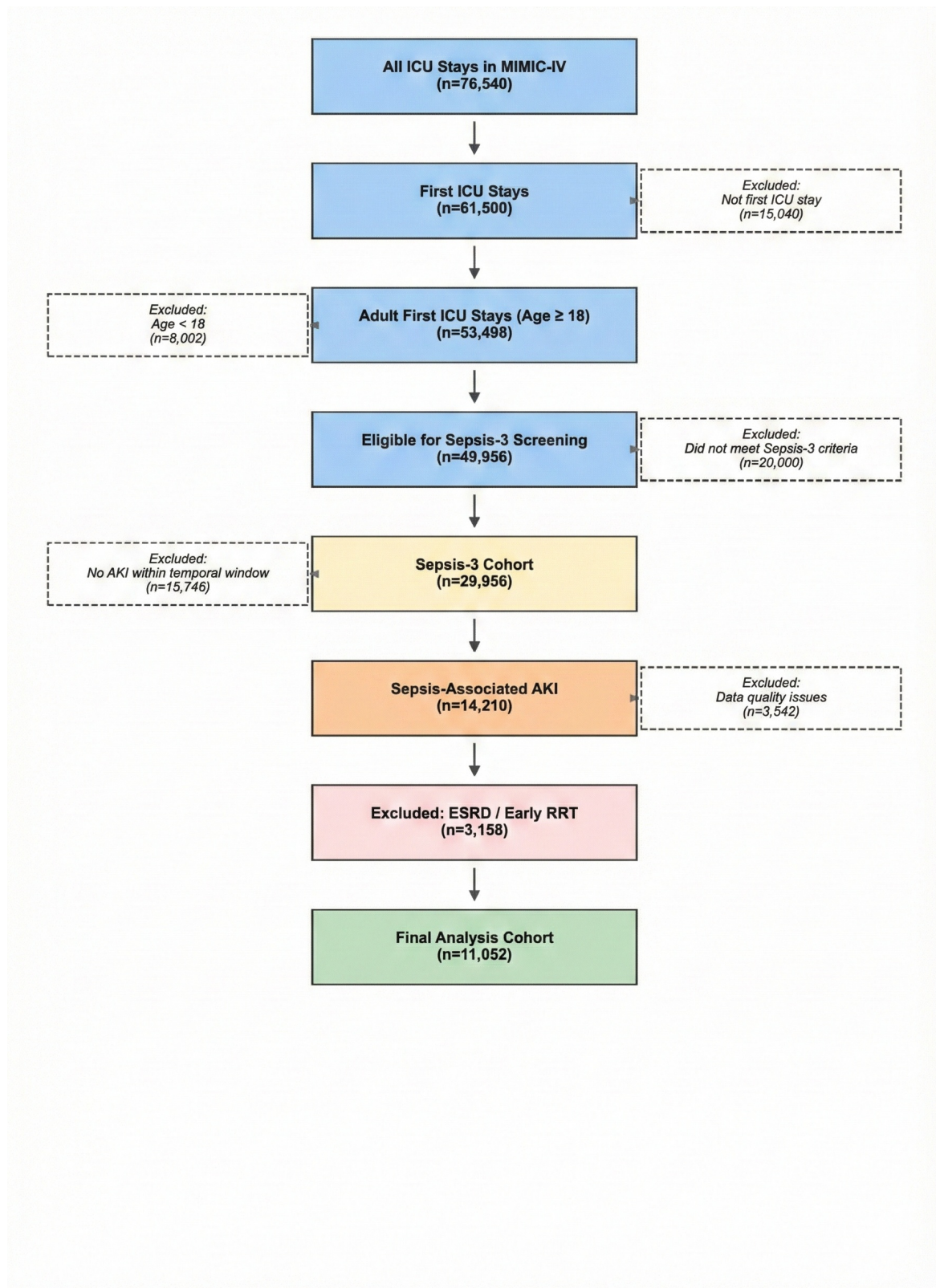
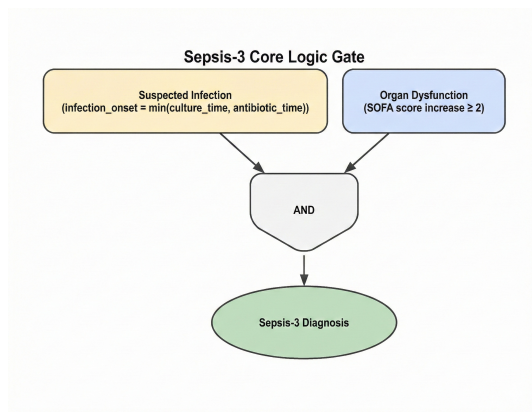
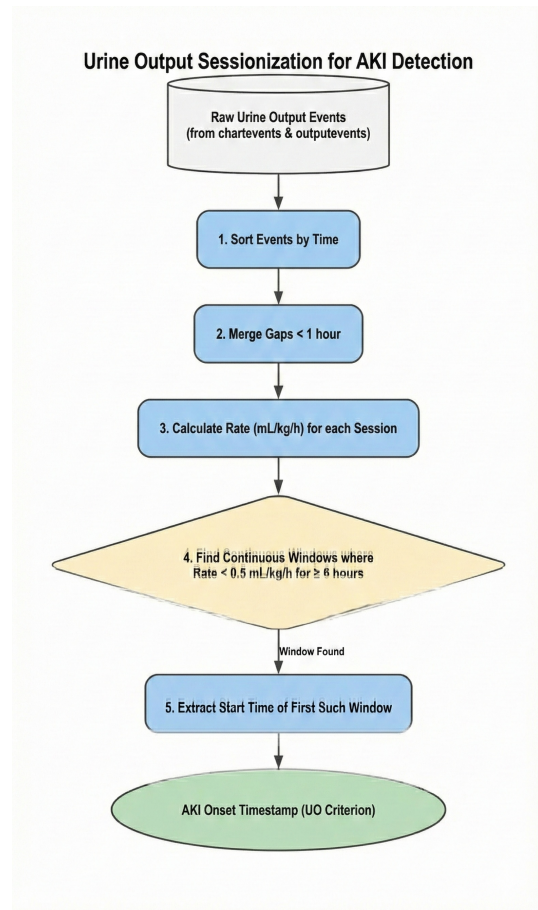


Figure 5.3: CONSORT-style flow of cohort construction with key exclusions.



(a) Sepsis-3 core logic gate.



(b) UO sessionization for KDIGO (continuous windows <0.5 mL/kg/h for  $\geq 6$  h).

Figure 5.4: Clinical rule components used for cohort labeling.



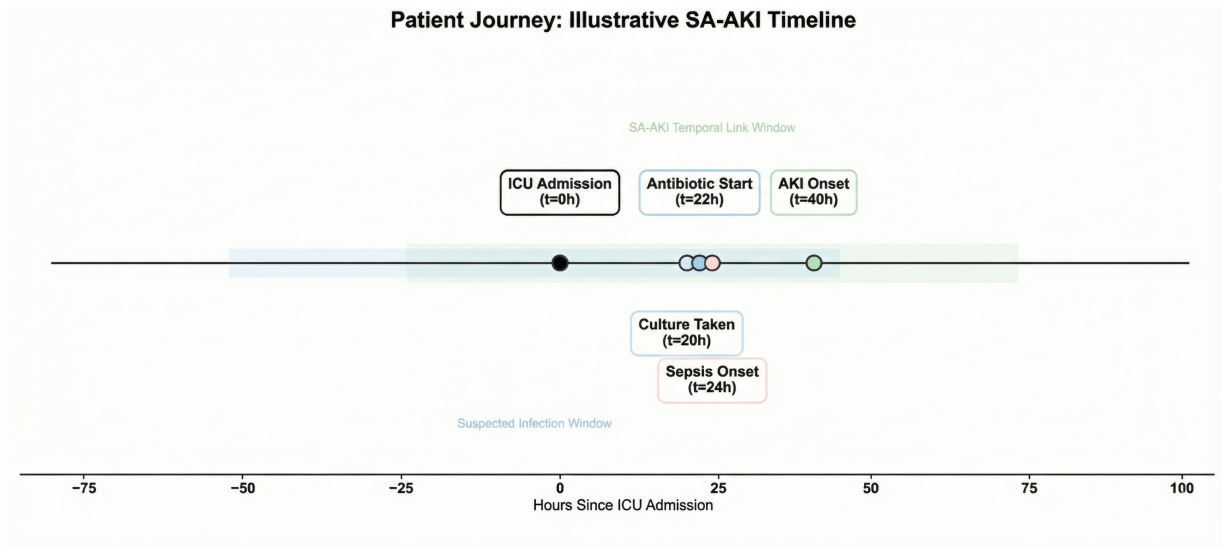


Figure 5.5: Illustrative patient journey with T0–T24 feature window, infection window, Sepsis onset, and AKI onset; the SA–AKI link is  $\pm 48$  h around Sepsis onset.

Table 5.1: Overall cohort snapshot (final build).

| Metric                                      | Value                       |
|---------------------------------------------|-----------------------------|
| In-hospital death                           | <b>26.7%</b> (2,684/10,036) |
| Male                                        | <b>53.0%</b> (5,322/10,036) |
| AKI trigger = Creatinine (vs. Urine Output) | <b>40.7%</b> (4,087/10,036) |

Table 5.2: Therapy/exposure prevalence in the first 24 h (1/0 encoding).

| Day-1 therapy / exposure flag         | Prevalence                  |
|---------------------------------------|-----------------------------|
| Mechanical ventilation within 24 h    | <b>53.0%</b> (5,320/10,036) |
| Vasopressor within 24 h               | <b>36.6%</b> (3,676/10,036) |
| Renal replacement therapy within 24 h | <b>69.0%</b> (6,922/10,036) |

## 5.7 EXPLORATORY ANALYSIS AND PHENOTYPING OF THE TRAINING COHORT

### 5.7.1 Statistical Validation of Data Integrity

A foundational step of the EDA was to statistically validate the cohort and the train-test split. As shown in Figure 5.6, key clinical variables exhibited highly significant differences between survivor and non-survivor groups within the training set (all  $p < 0.001$ , Mann-

Whitney U test), suggesting that the selected features contain a strong prognostic signal. The same tests revealed no significant distributional differences for these variables between the training and test sets (all  $p > 0.05$ ), indicating that the stratified split was successful and that the test set may be regarded as a representative sample for evaluating final model performance.

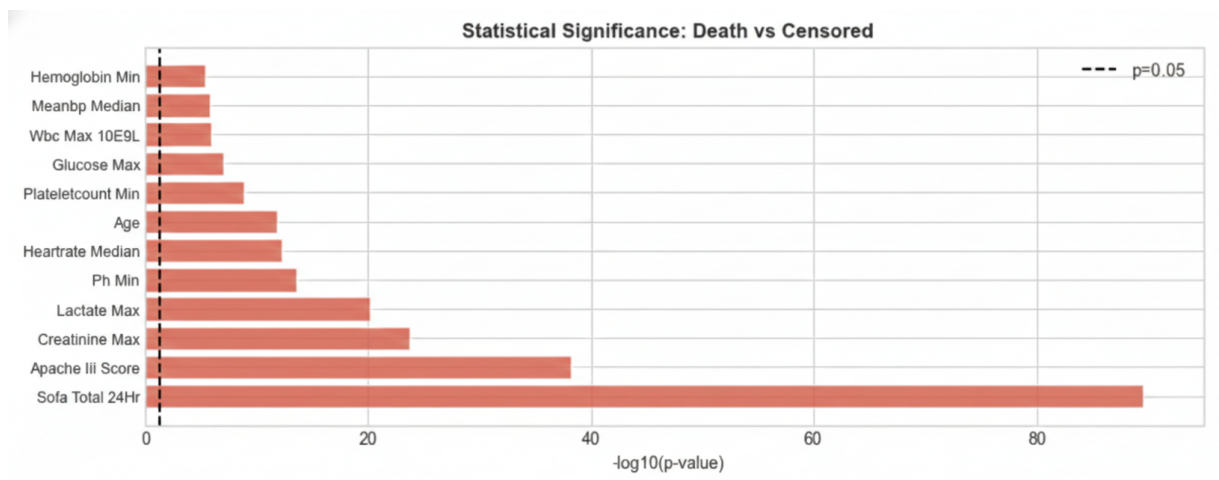


Figure 5.6: Statistical significance of key clinical variables between survivor and non-survivor groups, confirming that the selected features carry a strong prognostic signal ( $p < 0.001$ , Mann-Whitney U test).

### 5.7.2 Analysis of Missing Data Patterns

The pattern of missing data was investigated as a potential source of information. Of the 161 initial features, 101 had missing values. Little's MCAR test strongly rejected the null hypothesis ( $p \approx 0.0$ ), indicating the data were not Missing Completely At Random. A subsequent chi-square analysis revealed that the absence of 46 specific measurements was significantly associated with mortality ( $p < 0.05$ ). The ten most significant associations are shown in Table 5.3, highlighting that the absence of liver function and tissue injury tests was highly prognostic. These findings suggest that patterns of clinical data collection are themselves informative predictors of outcome.

Table 5.3: Top 10 most significant associations between missingness indicators and in-hospital mortality.

| Indicator              | Chi-Square | p-value | Cramér's V |
|------------------------|------------|---------|------------|
| ast_n_missing          | 166.46     | < 0.001 | 0.144      |
| ast_max_missing        | 166.46     | < 0.001 | 0.144      |
| alkphos_median_missing | 165.27     | < 0.001 | 0.143      |
| alkphos_n_missing      | 165.27     | < 0.001 | 0.143      |
| tbili_n_missing        | 163.90     | < 0.001 | 0.143      |
| tbili_max_mgdl_missing | 163.90     | < 0.001 | 0.143      |
| ldh_max_ul_missing     | 96.66      | < 0.001 | 0.110      |
| ldh_n_missing          | 96.66      | < 0.001 | 0.110      |
| calcium_n_missing      | 73.11      | < 0.001 | 0.095      |
| calcium_min_missing    | 73.11      | < 0.001 | 0.095      |

### 5.7.3 Clinical Phenotypes of Survivor vs. Non-Survivor Groups

The EDA revealed distinct clinical phenotypes between survivors and non-survivors in the training set. As illustrated in Figure 5.7, non-survivors were significantly older and had higher baseline severity of illness, reflected by higher APACHE III and SOFA scores. Physiologically, the non-survivor phenotype was characterized by more severe multi-organ dysfunction, including worse renal function (higher maximum creatinine) and greater metabolic distress (higher maximum lactate).

This phenotyping was further supported by correlation analysis (Figure 5.8). The overall SOFA score (`sofa_total_24hr`) had the strongest positive correlation with death (0.234), reinforcing its role as a global marker of risk. Other features strongly associated with mortality included markers of hematological dysfunction (`rdw_median_pct`, 0.188) and hepatic injury (`sofa_hepatic_24hr`, 0.169). In contrast, variables indicating physiological stability and adequate organ perfusion, such as total urine output (`urine_output_total_24hr`, -0.113) and minimum pH (`ph_min`, -0.101), were the strongest predictors of survival. Together, these findings provide a data-driven characterization of the high-risk SA-AKI patient and directly informed the feature set for supervised modeling.

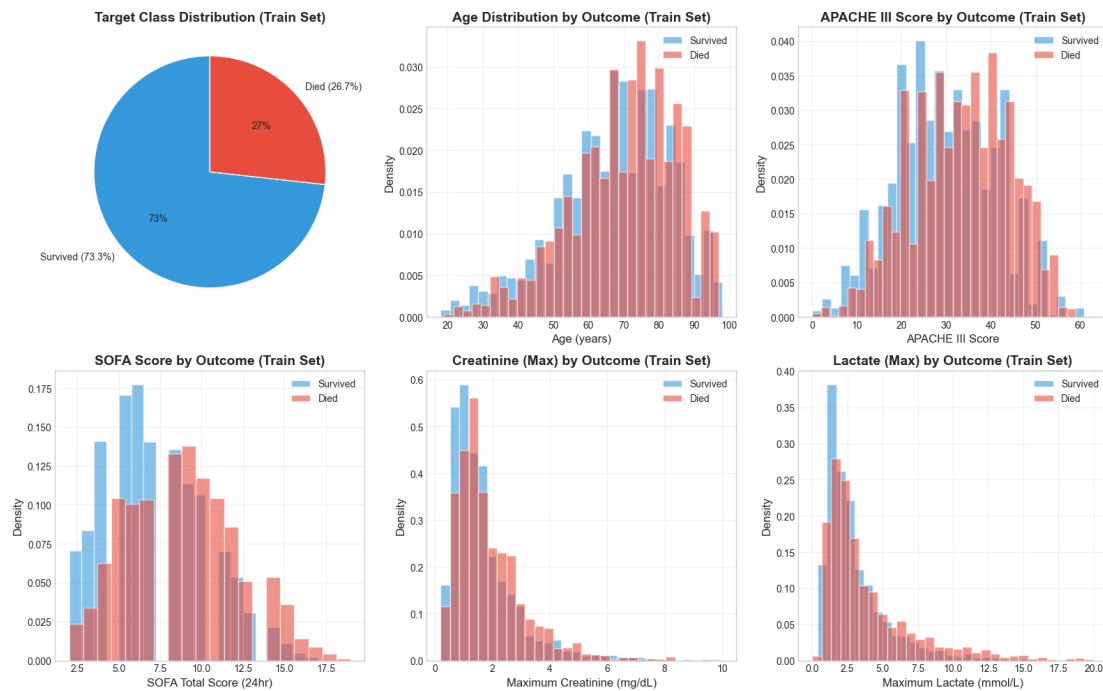


Figure 5.7: Distributions of key clinical variables in the training set, stratified by mortality outcome. Non-survivors consistently show distributions shifted towards higher severity.

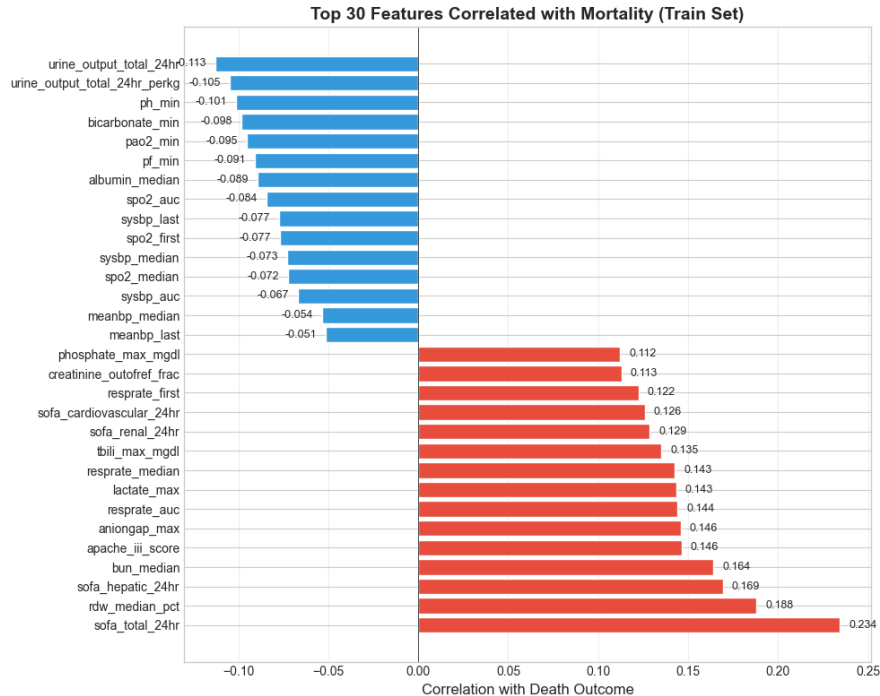


Figure 5.8: Top 30 features with the strongest positive (red) and negative (blue) correlation with in-hospital mortality in the training set.



# CHAPTER 6

## RESULTS

This chapter presents the outcomes of the predictive modeling pipeline. It begins by detailing the results of the extensive model benchmarking process, leading to the selection and final evaluation of the primary classification model. Subsequently, it presents the performance and interpretation of the time-to-event models. All results were generated from the held-out test set, following the strict, leakage-aware methodology established in the preceding chapters.

### 6.1 SUPERVISED MORTALITY PREDICTION

The primary objective of this thesis was the development of a binary classifier for in-hospital mortality. This was pursued through a two-stage process: benchmarking of various algorithms, followed by an in-depth evaluation of the final selected model.

#### 6.1.1 Model Benchmarking and Selection

A wide range of classification algorithms was systematically evaluated to identify the most suitable candidate for the clinical task. The benchmark included traditional statistical models (Logistic Regression, LDA), regularized linear models (Ridge, Lasso, ElasticNet), and more complex, non-linear ensembles (Random Forest, XGBoost, LightGBM, DNN++). To address the inherent class imbalance in the dataset, two primary strategies were compared: intrinsic class weighting and various oversampling techniques (e.g., SMOTE, BorderlineSMOTE).

The results of this benchmark are summarized in Table 6.1. A clear trend emerged: tree-based ensembles consistently outperformed linear models in terms of discrimination, as measured by the ROC-AUC. The **LightGBM model trained with class weights** was

identified as the top-performing model, achieving the highest ROC-AUC of 0.749 in the initial validation. While oversampling methods tended to produce higher F1-scores, this was often at the expense of precision and did not yield a superior ROC-AUC. Given that the primary goal was to accurately discriminate high-risk from low-risk patients across all decision thresholds, ROC-AUC was prioritized, and the LightGBM model was selected for the final stage of tuning, calibration, and evaluation.

Table 6.1: Best performance by algorithm type from the benchmarking experiments, sorted by ROC-AUC.

| Model Algorithm                    | ROC-AUC | F1-Score |
|------------------------------------|---------|----------|
| LightGBM                           | 0.7492  | 0.4129   |
| XGBoost                            | 0.7444  | 0.4527   |
| Logistic Regression                | 0.7382  | 0.5227   |
| Linear Discriminant Analysis (LDA) | 0.7364  | 0.3756   |
| ElasticNet                         | 0.7363  | 0.5185   |
| Ridge                              | 0.7349  | 0.5174   |
| Deep Neural Network (DNN++)        | 0.7347  | 0.4471   |
| Lasso                              | 0.7331  | 0.5244   |
| Random Forest                      | 0.7137  | 0.3468   |

### 6.1.2 Final Model Performance on the Test Set

The selected LightGBM model was subjected to a final evaluation pipeline. After hyperparameter tuning with Optuna, the model’s raw outputs were calibrated using isotonic regression on a validation split of the training data. The optimal classification threshold was determined to be 0.2704 by maximizing Youden’s J statistic on this same validation set.

The final, calibrated model was then applied to the unseen held-out test set. The model achieved a ROC-AUC of 0.749, which was taken as evidence of strong and generalizable discriminative ability. The complete performance metrics are detailed in Table 6.2. The confusion matrix (Figure 6.1a) provides a clear picture of the model’s performance at the chosen clinical threshold: it correctly identified 368 of the 537 patients who died (a recall of 0.685) while maintaining a precision of 0.422. The ROC curve (Figure 6.1b)

further illustrates the model’s consistent performance across the full range of decision thresholds.

Table 6.2: Performance of the final calibrated LightGBM model on the held-out test set.

| Metric               | Value  |
|----------------------|--------|
| ROC-AUC              | 0.7488 |
| Accuracy             | 0.6648 |
| Precision            | 0.4220 |
| Recall (Sensitivity) | 0.6853 |
| F1-Score             | 0.5224 |

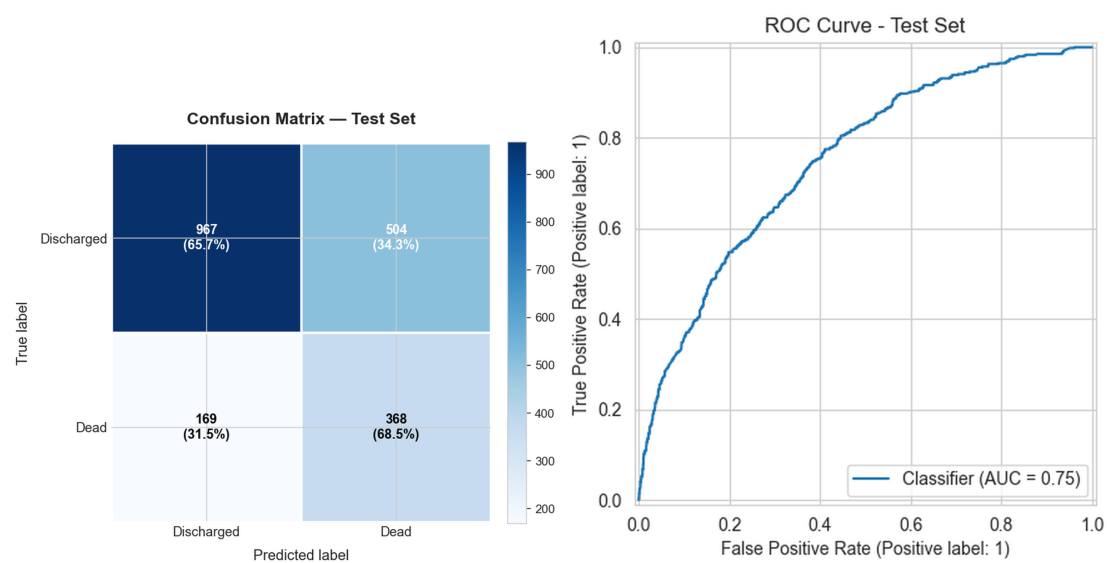


Figure 6.1: (a) Confusion matrix of the final model on the test set at the optimal threshold (0.2704). (b) ROC curve for the final model on the test set, showing an AUC of 0.75.

### 6.1.3 Model Interpretability with SHAP

To ensure the model’s clinical relevance and transparency, its predictions were interpreted using SHAP. The beeswarm plot in Figure 6.2 provides a global summary of the features driving the model’s output. This analysis suggests that the model’s logic aligns with established clinical knowledge.

The most influential feature was the `sofa_total_24hr` score, a global measure of organ dysfunction. Other key predictors of mortality included markers of hematological

dysfunction (rdw\_median\_pct), overall illness severity (apache\_iii\_score), advanced age, and renal failure (bun\_median). Conversely, higher urine output (urine\_output\_total\_24hr), a sign of preserved kidney function, was a strong predictor of survival. This concordance between the model’s learned patterns and clinical pathophysiology has been interpreted as evidence supporting its validity.

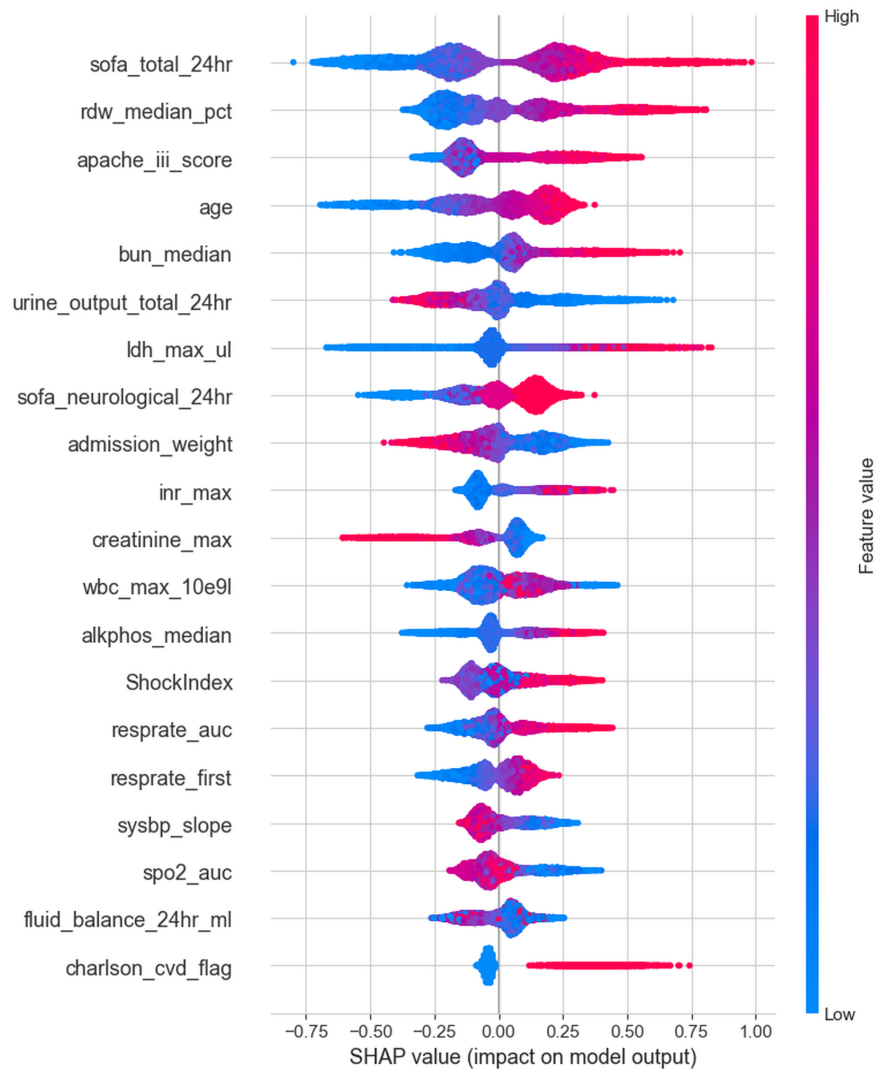


Figure 6.2: SHAP beeswarm summary plot for the top 20 features. Each dot represents a patient. The feature’s value is indicated by color (red=high, blue=low), and its impact on the mortality prediction is shown on the x-axis.

#### **6.1.4 Why LightGBM Outperforms Other Model Classes**

The selection of LightGBM as the final model was not arbitrary but was driven by its consistent superiority in the benchmark (Table 6.1). Several properties of the algorithm have been identified as contributing to this advantage in the context of SA–AKI mortality prediction.

LightGBM employs a leaf-wise tree growth strategy rather than the level-wise approach adopted by traditional gradient boosting implementations. This design concentrates splitting effort on the leaves that yield the largest reduction in loss, a property that has been found to be particularly effective when the feature space contains a mixture of continuous biomarkers with skewed distributions and sparse binary indicators such as comorbidity flags and missingness indicators. In addition, LightGBM natively captures the interaction structure that characterizes clinical data. SA–AKI mortality is not driven by any single variable in isolation but rather by the joint configuration of organ dysfunction scores, hemodynamic instability, and metabolic derangement. Tree ensembles represent these higher-order interactions without requiring explicit feature engineering, whereas linear models (Logistic Regression, Ridge, Lasso, ElasticNet) are constrained to additive main effects unless interactions are manually specified. This distinction is reflected in the benchmark: the best-performing linear model (Logistic Regression, AUROC 0.738) trailed LightGBM (AUROC 0.749) by a margin that, while modest in absolute terms, was observed to be consistent across cross-validation folds and to correspond to meaningful reclassification of patients near the decision boundary.

The handling of class imbalance further differentiated LightGBM from alternative approaches. The built-in `class_weight='balanced'` mechanism, which reweights the loss function inversely proportional to class frequency, was found to be more effective than synthetic oversampling (SMOTE, ADASYN) for this dataset. Oversampling methods tended to inflate recall at the cost of precision without improving overall discrimination, a pattern consistent with the known risks of synthesizing examples in overlapped

regions of the feature space (Blagus and Lusa, 2013; Sáez *et al.*, 2015). The Optuna-based hyperparameter search explored a richer space than grid search, encompassing regularization parameters (`reg_alpha`, `reg_lambda`), tree depth, minimum child samples, and feature and bagging fractions, which collectively controlled overfitting and contributed to improved generalization on the held-out test set.

### **6.1.5 Evidence for Trustworthiness and Clinical Credibility**

A clinician evaluating a risk score requires more than a single discrimination metric; the credibility of the model must be established through multiple complementary lines of evidence. The trustworthiness of the predictions presented here has been assessed along five dimensions.

The first dimension concerns methodological integrity. The AUROC of 0.749 was obtained under conditions specifically designed to prevent optimistic bias: all preprocessing was fitted exclusively on training data, no post-T24 information entered the feature set, the test set was used exactly once for final evaluation, and no feature selection or imputation step was permitted access to test-set outcomes. In contrast, studies reporting AUROC values of 0.79–0.88 for SA–AKI mortality have been found to include length of stay as a predictor (Chen *et al.*, 2025), to perform feature selection on the full dataset (Li *et al.*, 2023), or to employ whole-stay aggregation windows that embed future information. When these methodological advantages are removed, the achievable discrimination for a genuine early warning system would be expected to be lower, and the result obtained here is consistent with that expectation.

The second dimension relates to calibration. The predicted probabilities were explicitly calibrated using isotonic regression on a validation split, ensuring that when the model assigns a 30% risk of death, approximately 30% of patients in that probability bin were observed to have died. Calibration is essential for clinical utility because treatment decisions depend on the absolute magnitude of risk rather than merely on the ranking

of patients. It has been noted that several prior studies did not report calibration at all (Chen *et al.*, 2025; Li *et al.*, 2023).

The third dimension is clinical face validity, as assessed through SHAP analysis. The SHAP analysis (Figure 6.2) confirmed that the top predictors identified by the model (SOFA score, RDW, APACHE III, age, BUN, and urine output) are established clinical risk factors for ICU mortality. A model whose predictions are driven by clinically interpretable signals has been considered more likely to generalize than one that exploits dataset-specific artifacts.

The fourth dimension involves decision-curve analysis. The survival models were shown to demonstrate positive net benefit over the “treat all” and “treat none” strategies across a clinically relevant range of risk thresholds (8–30%), providing direct evidence that the use of the model to inform decisions would be expected to improve outcomes compared with default strategies (Figure 6.7).

The fifth dimension is the concordance observed across fundamentally different model families. The consistency of results across LightGBM (AUROC 0.749), CoxPH (C-index 0.693), and DeepSurv (C-index 0.688) provides a form of triangulation. If the predictive signal were spurious or driven by leakage, it would be unlikely to manifest consistently across a gradient-boosted classifier, a semi-parametric survival model, and a neural survival network.

#### **6.1.6 Comparison with Prior Work and the Case for a Realistic Benchmark**

Recent studies on SA–AKI mortality using MIMIC–IV have reported ROC–AUC values ranging from 0.79 to 0.88 (Chen *et al.*, 2025; Li *et al.*, 2023; Nong *et al.*, 2024). However, as discussed in Chapter 2, many of these results are likely to represent overestimations arising from methodological shortcomings: information leakage through whole-stay data aggregation, inclusion of length of stay as a predictor, global fitting of preprocessing steps, and the absence of calibration. The key methodological differences are summarized in

Table 6.3.

Table 6.3: Methodological comparison with representative prior SA–AKI mortality studies.

| Criterion                | This work | Chen (2025) | Li (2023) | Hu (2021)  |
|--------------------------|-----------|-------------|-----------|------------|
| Feature window           | T0–T24    | Unspecified | T0–T24    | Whole-stay |
| LOS as predictor         | No        | Yes         | No        | No         |
| Train-only preprocessing | Yes       | No          | No        | No         |
| Missingness indicators   | Yes       | No          | No        | No         |
| Explicit calibration     | Yes       | No          | No        | No         |
| Separate validation set  | Yes       | No          | No        | No         |
| Reported AUROC           | 0.749     | 0.878       | 0.794     | —          |
| Reported C-index         | 0.69      | —           | —         | 0.72       |

The AUROC of 0.749 should therefore be interpreted not as an inferior result but as a more realistic and credible estimate of what is achievable for a genuine early warning system that relies exclusively on information available at the bedside within 24 hours of ICU admission. This work has been positioned as a methodologically sound benchmark against which future studies may be compared on equal footing.

#### 6.1.7 Limitations: External Validation and Dataset Scope

A key limitation of this work, shared with the majority of SA–AKI prediction studies, is the absence of external validation on an independent dataset. The model was developed and evaluated exclusively on MIMIC–IV, which reflects the patient population, clinical practices, and documentation patterns of a single tertiary-care center (Beth Israel Deaconess Medical Center, Boston). Performance drift across hospitals has been documented in the ICU prediction literature (Rockenschaub *et al.*, 2024), and the degree to which the model would generalize to other institutions remains an open empirical question.

The most natural candidate for external validation has been identified as the eICU Collaborative Research Database, a multi-center dataset covering over 200 hospitals across the United States. However, eICU presents several harmonization challenges that



would need to be addressed: differences in laboratory item coding, variable availability (certain biomarkers may be less consistently recorded across sites), and population-level differences in case mix and care intensity. A careful external validation would require re-application of the same cohort definition (Sepsis-3 + KDIGO), re-derivation of T0–T24 features from the eICU schema, and evaluation of the frozen model without retraining. This has been identified as the immediate next step in Chapter 7.

Until external validation has been completed, the model’s predictions should be interpreted as internally valid risk estimates specific to the MIMIC-IV population. Recalibration on the target population would be expected to be necessary before the model could be considered for bedside decision support at any external institution. It has been noted that many prior studies have claimed broad generalizability without providing external evidence to support such claims (Chen *et al.*, 2025), and the explicit acknowledgment of this limitation in the present work has been intended to encourage more transparent reporting practices in the field.

## **6.2 TIME-TO-EVENT (SURVIVAL) ANALYSIS**

To provide a more dynamic assessment of patient risk, survival models were developed to predict the time to in-hospital death using the same T0–T24 feature set.

### **6.2.1 Cox Proportional Hazards Model**

After feature selection and penalization tuning, the final Cox Proportional Hazards (CoxPH) model achieved a test C-index of 0.693, indicating good discriminative ability over time. The forest plot in Figure 6.3 reveals the key drivers of hazard. Comorbidities like metastatic cancer and high SOFA scores were the strongest predictors of an increased hazard of death. Of note, early interventions such as RRT and vasopressor use were associated with a reduced hazard, a finding that is likely attributable to confounding by indication, whereby the sickest patients receive the most aggressive interventions, rather than to a causal protective effect of the therapies themselves. The model’s ability to

effectively stratify patients into distinct risk groups was confirmed by the clear separation of Kaplan-Meier curves for its predicted risk quartiles (Figure 6.4, log-rank  $p < 0.001$ ).

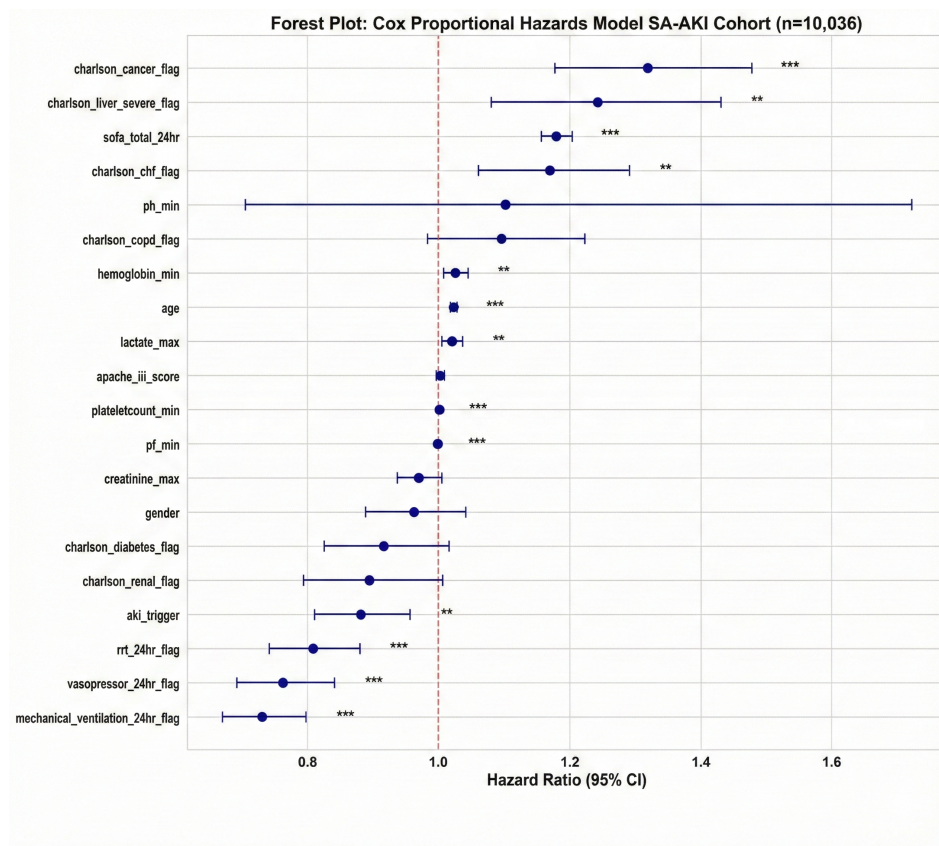


Figure 6.3: Forest plot of hazard ratios for the top 20 predictors in the final CoxPH model. Values to the right of the dashed line ( $HR > 1$ ) indicate an increased risk of death.

## 6.2.2 DeepSurv Model

A custom neural network-based survival model, DeepSurv, was also developed and achieved a comparable test C-index of 0.688. Like the Cox model, it demonstrated strong risk stratification (Figure 6.5, log-rank  $p < 0.001$ ). The clinical utility of the DeepSurv model was further interrogated through calibration and decision curve analysis (DCA). The calibration plots (Figure 6.6) showed good agreement between predicted and observed event probabilities at key clinical horizons, although with a slight tendency to overestimate risk in the highest-risk bins, a pattern that could be addressed in future work with more advanced calibration techniques. The DCA (Figure 6.7) demonstrated that

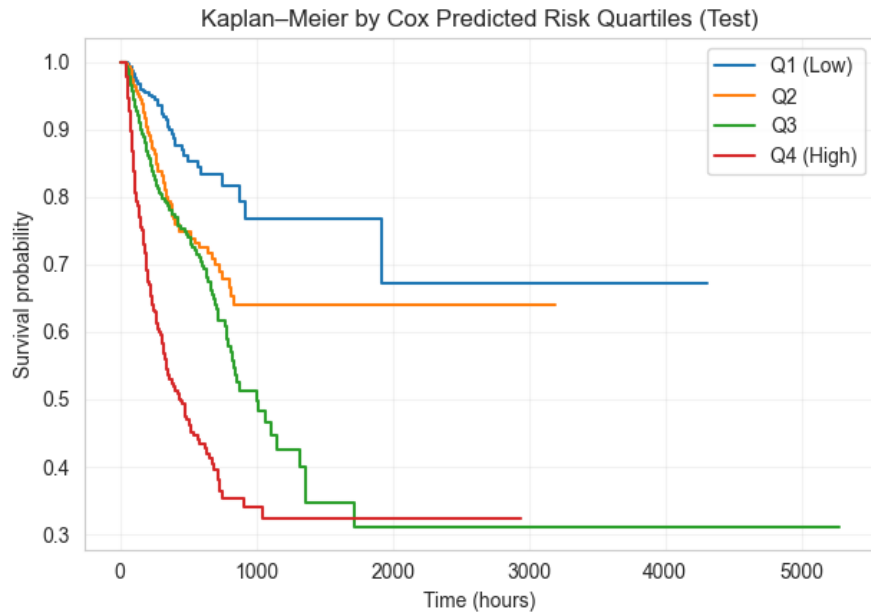


Figure 6.4: Kaplan-Meier survival curves for the test set, stratified by risk quartiles predicted by the CoxPH model, demonstrating effective risk stratification.

the use of the model to inform clinical decisions provides a net benefit over the default strategies of treating all or no patients for risk thresholds between 8% and 30%.

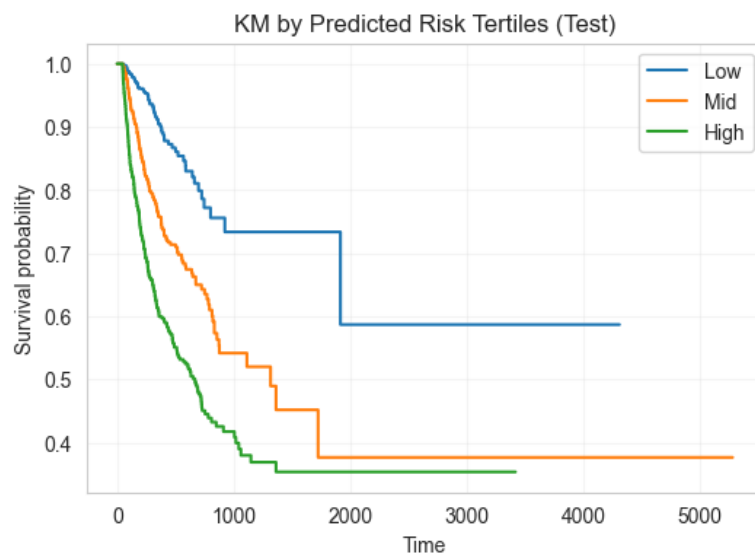


Figure 6.5: Kaplan-Meier survival curves for the test set, stratified by risk tertiles predicted by the DeepSurv model.

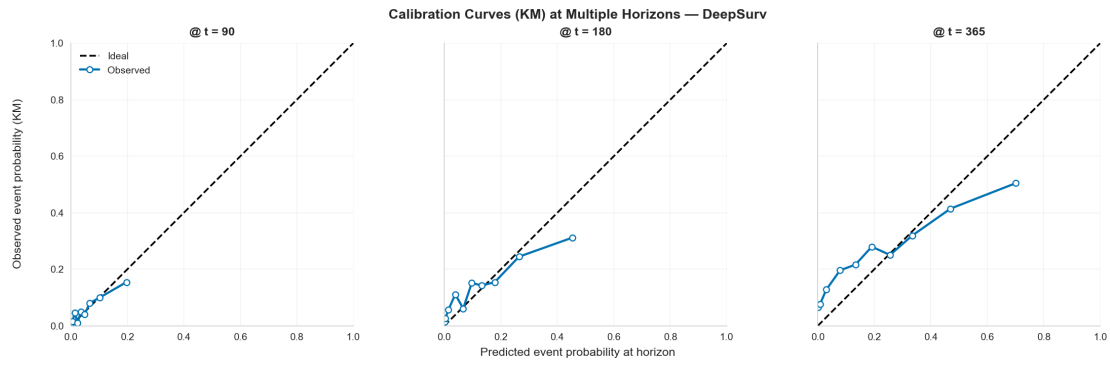


Figure 6.6: Calibration plots for the DeepSurv model at 90, 180, and 365 hours.

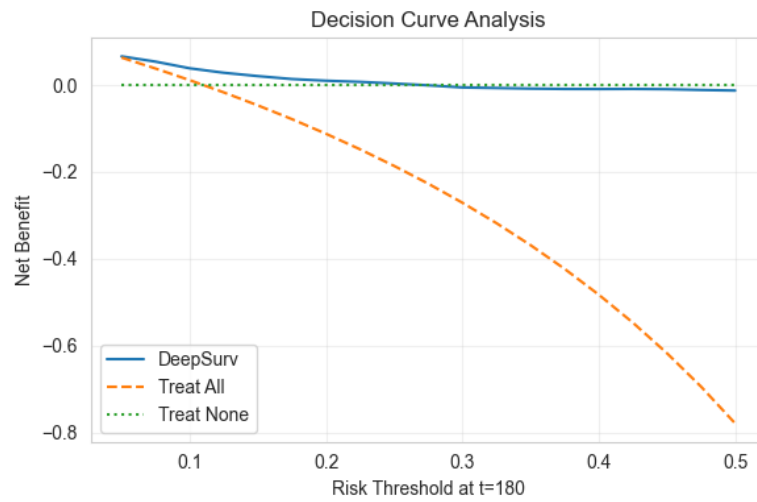


Figure 6.7: Decision Curve Analysis for the DeepSurv model at a 180-hour horizon, demonstrating positive net benefit across a range of clinically relevant risk thresholds.

### 6.2.3 Synthesis of Survival Analysis

Both the CoxPH and DeepSurv models delivered consistent and clinically interpretable predictions of time-to-death, with C-indices of 0.693 and 0.688, respectively. These results are comparable to those from other methodologically sound survival studies in critical care. For instance, Hu *et al.* (2021) reported a C-index of 0.72 but used whole-stay aggregated features, which introduces a high risk of data leakage. The results obtained in the present work, derived strictly from T0–T24 data, therefore provide a more reliable and clinically actionable benchmark for early dynamic risk stratification in the SA–AKI

population.

# CHAPTER 7

## CONCLUSIONS AND FUTURE WORK

### 7.1 SUMMARY OF WORK

This thesis has presented the design, implementation, and validation of a machine learning framework for early mortality prediction in Sepsis-Associated Acute Kidney Injury (SA-AKI), with the aim of establishing a methodologically sound benchmark for the field. Through strict adherence to a 24-hour clinical window and the TRIPOD reporting guidelines, a transparent and reproducible pipeline has been developed that addresses the critical shortcomings of data leakage and performance overestimation that have been identified as prevalent in much of the existing literature.

The key contributions and findings are summarized as follows:

1. **High-Fidelity Cohort Construction:** A well-defined SA-AKI cohort of 10,036 patients was constructed from the MIMIC-IV database, underpinned by an ontology-integrated PostgreSQL warehouse that ensured reproducible definitions and data harmonization.
2. **Leakage-Aware Preprocessing:** A strict 80/20 train-test split served as the foundation for a preprocessing pipeline that included a formal analysis of the missing data mechanism, the creation and statistical selection of significant missingness indicators as novel features, and the train-only fitting of all data transformers to prevent information leakage.
3. **Interpretable Predictive Modeling:** Following a benchmark of machine learning algorithms, a LightGBM model was selected, tuned, and calibrated. On the held-out test set, the final model achieved an AUROC of 0.749 and an F1-score of 0.522. The clinical validity of the model was supported by SHAP analysis, which revealed that predictions were driven by established risk factors such as SOFA score, RDW, and age.
4. **Dynamic Risk Assessment with Survival Analysis:** The utility of the early-window features was extended to time-to-event modeling. Both a Cox Proportional Hazards model and a custom DeepSurv model were developed, each achieving a C-index of approximately 0.69 on the test set. These models were found to be effective at stratifying patients into distinct risk groups over time, providing a more

dynamic view of clinical risk.

Collectively, this work has provided a full pipeline that yields a clinically interpretable predictive model while maintaining the methodological discipline that has been argued to be necessary for building clinical decision support tools suitable for consideration in real-world settings.

## 7.2 FUTURE WORK

Building upon the foundation established in this thesis, several avenues for future research have been identified:

- **External Validation and Transportability:** The immediate next step would be to evaluate the model’s performance on external datasets, with the eICU Collaborative Research Database representing the most natural candidate. The eICU database, covering over 200 hospitals across the United States, would provide a demanding test of generalizability across different patient populations, clinical practices, and documentation standards. A validation protocol would involve re-applying the Sepsis-3 and KDIGO cohort definitions to eICU, re-deriving T0–T24 features from its schema, and evaluating the frozen model without retraining. Should performance degradation be observed, recalibration on the target population would be explored before any clinical deployment could be considered.
- **Monte Carlo Simulation for Synthetic Data Augmentation:** The use of Monte Carlo methods to generate synthetic patient trajectories that respect the joint distribution of clinical variables observed in the training data represents a promising direction. Copula-based simulation or parametric bootstrap approaches could be employed to sample realistic feature vectors from the estimated multivariate distribution of T0–T24 measurements, preserving the correlation structure among biomarkers, vital signs, and severity scores. Such an approach would serve two purposes: augmenting the training set to improve model stability for underrepresented subgroups, and enabling sensitivity analyses to assess how model performance varies under perturbations of the input distribution. Unlike SMOTE or ADASYN, which operate in feature space without regard to clinical plausibility, a Monte Carlo approach grounded in domain-specific distributional assumptions would be expected to generate physiologically coherent synthetic patients (Blagus and Lusa, 2013). Generative adversarial networks and variational autoencoders trained on the MIMIC-IV cohort represent more flexible alternatives, though at the cost of reduced interpretability of the generation process.
- **Multi-Database Risk Framing:** To strengthen the evidence base for the degree of risk within the first 24 hours, future work should seek to integrate data from

multiple sources beyond MIMIC–IV. The Amsterdam UMCdb, the HiRID dataset from the Bern University Hospital, and the ANZICS Adult Patient Database each offer complementary patient populations and clinical practices. A federated or meta-analytic approach that estimates risk across these databases would provide stronger evidence for the universality of the 24-hour risk signal and would help disentangle center-specific effects from genuine physiological predictors.

- **Exploration of Different Temporal Windows:** While this study focused on a 24-hour window, the predictive power of earlier windows (e.g., 6 or 12 hours) could be investigated to facilitate more timely interventions, or later windows (e.g., 48 hours) could be examined to capture evolving patient trajectories.
- **Model Fairness and Equity Audit:** An important step before any clinical deployment would be to audit the model’s performance across different demographic subgroups (e.g., by sex, age, and ethnicity) to verify that predictions are equitable and to identify and mitigate any potential biases that may have been introduced through the training data.
- **Prospective Clinical Impact Study:** The ultimate goal would be to translate this research into clinical practice. A prospective, observational study could be designed to deploy the model in a real-world ICU setting as a decision support tool, allowing for the evaluation of its impact on clinical workflows, decision-making, and ultimately, patient outcomes.



# APPENDIX A

## CLINICALLY PLAUSIBLE VALUES

This appendix lists clinically plausible (extreme–but–possible) minima and maxima for numeric features in the sepsis–AKI ICU cohort (first 24 h), intended as *winsorization* bounds during preprocessing. Ranges reflect ICU/sepsis/AKI physiology rather than healthy outpatients and are aligned with the unit-harmonized guardrails used in the curated dataset.

### DEMOGRAPHICS

| Variable         | Min | Max | Units |
|------------------|-----|-----|-------|
| Age              | 18  | 120 | years |
| Admission weight | 30  | 300 | kg    |

*Notes.* Adult ICUs typically include patients  $\geq 18$  y; extreme body weights spanning severe malnutrition to massive obesity motivate 30–300 kg as a pragmatic span.

### TIMING ANCHORS (DESCRIPTIVE ONLY)

| Variable                            | Min | Max   | Units |
|-------------------------------------|-----|-------|-------|
| Time to death                       | 0   | 2,880 | h     |
| Time to AKI onset                   | 0   | 168   | h     |
| Time to sepsis onset                | 0   | 72    | h     |
| Time to mechanical ventilator start | 0   | 120   | h     |
| Time to vasopressor start           | 0   | 120   | h     |

*Notes.* Onset times cluster near ICU admission in sepsis and AKI; the upper bounds allow delayed recognition/therapy within the early ICU course.

## VITAL SIGNS (FIRST 24 H)

| Variable                 | Min | Max | Units       |
|--------------------------|-----|-----|-------------|
| Heart rate               | 20  | 250 | beats/min   |
| Systolic blood pressure  | 50  | 300 | mmHg        |
| Diastolic blood pressure | 30  | 200 | mmHg        |
| Mean arterial pressure   | 30  | 200 | mmHg        |
| Respiratory rate         | 4   | 50  | breaths/min |
| Oxygen saturation        | 50  | 100 | %           |

*Notes.* Sustained tachyarrhythmias can transiently reach 200–250 bpm (Cleveland Clinic 2024); severe shock and hypertensive crisis motivate BP limits (Guy’s and St Thomas’ Specialist Care 2023; Author and Others 2021). Profound hypoxemia may drive SpO<sub>2</sub> towards 50–60% while values >100% are non-physiologic (Mayo Clinic 2024b; Family Practice Notebook 2024).

## FLUID BALANCE AND URINE OUTPUT (FIRST 24 H)

| Variable                    | Min     | Max     | Units |
|-----------------------------|---------|---------|-------|
| Net fluid balance (24 h)    | −10,000 | +15,000 | mL    |
| Fluid balance per kg (24 h) | −100    | +200    | mL/kg |
| Urine output (24 h)         | 0       | 20,000  | mL    |
| Urine output per kg (24 h)  | 0       | 200     | mL/kg |

*Notes.* Bounds reflect aggressive resuscitation/diuresis and anuria–polyuria extremes early in critical illness.

## BLOOD GASES AND OXYGENATION (FIRST 24 H)

*Notes.* Reported survivals at pH near 6.5–6.8 are exceptional (Donadello *et al.* 2021); permissive hypercapnia and ventilatory failure bound PaCO<sub>2</sub> (Respiratory Therapy

| Variable                                       | Min | Max | Units |
|------------------------------------------------|-----|-----|-------|
| Arterial pH                                    | 6.5 | 7.7 | —     |
| PaCO <sub>2</sub>                              | 10  | 100 | mmHg  |
| PaO <sub>2</sub>                               | 30  | 500 | mmHg  |
| PF ratio (PaO <sub>2</sub> /FiO <sub>2</sub> ) | 0   | 600 | mmHg  |

Magazine 2020; Medscape 2024). On FiO<sub>2</sub>=1.0, PaO<sub>2</sub> can reach ~400–500 mmHg; PF ratios span 0–600 mmHg (Lévy *et al.* 2021; Family Practice Notebook 2024).

## ELECTROLYTES AND ACID–BASE (FIRST 24 H)

| Variable    | Min | Max | Units  |
|-------------|-----|-----|--------|
| Sodium      | 100 | 190 | mmol/L |
| Potassium   | 2.0 | 8.0 | mmol/L |
| Chloride    | 70  | 130 | mmol/L |
| Bicarbonate | 5   | 40  | mmol/L |
| Anion gap   | 3   | 30  | mmol/L |

*Notes.* Bounds cover high–gap metabolic acidosis and severe alkalosis; see Vasavada *et al.* (2023); The Royal Children’s Hospital Melbourne (2024) for context.

## RENAL FUNCTION AND GLYCEMIA (FIRST 24 H)

| Variable            | Min | Max  | Units |
|---------------------|-----|------|-------|
| Creatinine (serum)  | 0.3 | 15.0 | mg/dL |
| Blood urea nitrogen | 5   | 200  | mg/dL |
| Glucose (plasma)    | 20  | 800  | mg/dL |

*Notes.* Severe AKI/ESRD without dialysis can manifest with creatinine 10–15 mg/dL and BUN >100 mg/dL; hyperosmolar states often show glucose 600–800+ mg/dL (Apollo 24x7 2024; MedicalNewsToday 2018; Kitabchi *et al.* 2023).

### CALCIUM, MAGNESIUM, PHOSPHATE (FIRST 24 H)

| Variable          | Min | Max  | Units |
|-------------------|-----|------|-------|
| Calcium (total)   | 4.0 | 15.0 | mg/dL |
| Magnesium (serum) | 0.5 | 6.0  | mg/dL |
| Phosphate (serum) | 0.5 | 15.0 | mg/dL |

*Notes.* Life-threatening calcium extremes and clinically significant magnesium/phosphate derangements motivate these bounds (Merck Manuals Professional Edition 2023; Fong and Khan 2012).

### HEPATIC AND BILIARY PANEL (FIRST 24 H)

| Variable                   | Min | Max    | Units |
|----------------------------|-----|--------|-------|
| Alkaline phosphatase       | 20  | 1,200  | U/L   |
| AST                        | 5   | 10,000 | U/L   |
| ALT                        | 5   | 10,000 | U/L   |
| Total bilirubin            | 0   | 40     | mg/dL |
| Direct bilirubin           | 0   | 30     | mg/dL |
| Gamma-glutamyl transferase | 0   | 2,000  | U/L   |
| Albumin (serum)            | 1.0 | 6.0    | g/dL  |

*Notes.* Massive transaminase elevations in ischemic/toxic injury (Bernal and Wendon 2016; Lee 2008) and marked cholestatic patterns (Mayo Clinic 2024a) justify the upper bounds; severe hypoalbuminemia is common in critical illness (Nicholson *et al.* 2023).

### HEMATOLOGY AND COAGULATION (FIRST 24 H)

*Notes.* Extremes of anemia/polycythemia (Costa *et al.* 2019; Padmanabhan *et al.* 2020), critical thrombocytopenia and thrombocytosis (Blumberg *et al.* 2023; Wegmann *et al.*

| Variable               | Min | Max   | Units                  |
|------------------------|-----|-------|------------------------|
| Hemoglobin             | 3   | 20    | g/dL                   |
| Platelet count         | 5   | 1,000 | $\times 10^9/\text{L}$ |
| INR                    | 0.8 | 10.0  | ratio                  |
| aPTT                   | 15  | 200   | s                      |
| White blood cell count | 0   | 200   | $\times 10^9/\text{L}$ |

2020), and profound coagulopathy (Hylek *et al.* 1998) set these bounds. Leukocytosis may rarely exceed  $200 \times 10^9/\text{L}$ .

### INFLAMMATION AND TISSUE INJURY (FIRST 24 H)

| Variable              | Min | Max     | Units  |
|-----------------------|-----|---------|--------|
| C-reactive protein    | 0   | 500     | mg/L   |
| Lactate dehydrogenase | 50  | 10,000  | U/L    |
| Troponin (cardiac)    | 0   | 100     | ng/mL  |
| BNP or NT-proBNP      | 0   | 100,000 | pg/mL  |
| Lactate (serum)       | 0.5 | 20      | mmol/L |

*Notes.* Severe sepsis often elevates lactate; levels  $\geq 4$ –10 mmol/L indicate high risk (Levy and Evans 2021). Very high troponin reflects extensive myocardial injury.

### SEVERITY SCORES (DAY 1)

| Variable                   | Min | Max | Units  |
|----------------------------|-----|-----|--------|
| APACHE III score           | 0   | 299 | points |
| SOFA total (24 h)          | 0   | 24  | points |
| SOFA subscore (each organ) | 0   | 4   | points |

*Notes.* Ranges follow score definitions (Knaus *et al.* 1991; Physiopedia 2024).

**Implementation.** Bounds are applied after unit harmonization and basic plausibility checks; outliers beyond these limits are capped, not dropped. These limits match the clinical guardrails used to compute laboratory QC fractions in the main dataset.

## APPENDIX B

### FEATURE DICTIONARY AND DESCRIPTIVE TABLES

An annotated data dictionary for the sepsis-associated acute kidney injury (SA-AKI) cohort has been compiled to summarize the first 24 hours of intensive care and to support time-to-event analyses. The final build comprises  $n=10,036$  stays and  $p = 162$  features. All variables have been harmonized to target units before aggregation, and binary encodings have been standardized as indicated. Overall cohort characteristics were observed to include in-hospital mortality of 26.7

Endpoints and anchor times were defined as in Table B.1. The in-hospital death indicator was coded as **1=death**, 0=censored, with time measured from ICU admission to death or last known alive. Onset times for AKI (KDIGO), Sepsis-3, first invasive ventilation, and first vasopressor use were stored in hours from admission. Demographics included age (years) and a sex indicator (**1=male**, 0=female); harmonized admission weight (kg) was retained for indexing of fluid and urine features. The AKI trigger variable encoded the primary detection pathway (**1=creatinine**, 0=urine output).

Binary flags for therapies and day-1 exposures were defined as summarized in Table B.2 and encoded as 1/0. Charlson comorbidity components and the total points were represented as separate indicators (Table B.3), with component-wise prevalences reported for the full cohort.

Vital-sign bases and their canonical units were enumerated in Table B.4. Day-1 statistics were derived through a common suffix system (Table B.8), where *first*, *last*, *median*, *iqr*, *rng*, *delta*, *auc*, *slope*, and *n* reflected, respectively, the first/last value, 24 h median,

interquartile range, range, change (last–first), trapezoidal area–under–curve, per–hour linear slope, and observation count. For derivatives (*delta*, *slope*), a requirement of  $n \geq 2$  was imposed; otherwise, values were set to NULL while the companion count was preserved. The ShockIndex (HR/SBP) was stored as a dimensionless derived signal.

Fluid balance and urine output features were recorded for the initial 24 hours (Table B.5), with both absolute and weight–indexed forms (mL and mL/kg). Net fluid balance was calculated as total inputs minus outputs across all documented sources.

Laboratory and gas–exchange features were organized by base analyte with curated units and aggregates (Tables B.6–B.7). Unit normalization was performed *prior* to any quality control or statistical aggregation. Representative conversions included glucose mmol/L→mg/dL ( $\times 18$ ), albumin and hemoglobin g/L→g/dL ( $\div 10$ ), calcium mmol/L→mg/dL ( $\times 4.0$ ), chloride mg/dL→mmol/L ( $\div 3.545$ ), lactate mg/dL→mmol/L ( $\div 9$ ), blood gases kPa→mmHg ( $\times 7.5$ ), bilirubin  $\mu\text{mol/L}$ →mg/dL ( $\div 17.104$ ), magnesium mmol/L→mg/dL ( $\times 2.433$ ), phosphate mmol/L→mg/dL ( $\times 3.097$ ), troponin ng/L→ng/mL ( $\div 1000$ ), and BNP/NT–proBNP ng/L $\equiv$ pg/mL.

For each analyte, day–1 summaries were produced using count, median, minima, maxima, and, where specified, dynamics (*delta*) or terminal value (*last*). Quality–control fractions were retained when available: the share outside the reported reference interval (`_outofref_frac`), the share labeled abnormal by the lab system (`_abn_frac`), and the share of urgent orders (`*_stat_frac`).

All fractions were bounded to  $[0, 1]$  with the analyte–specific  $n$  as denominator. Clinically plausible guardrails were applied only for plausibility checks and out–of–range tallies (e.g., electrolytes, gases, bilirubin, hematology, CRP, LDH, troponin, BNP), rather than for winsorization.

Oxygenation was further characterized through a  $\text{PaO}_2/\text{FiO}_2$  ratio (PF).  $\text{FiO}_2$



measurements were normalized to a fraction within [0.21, 1.0] from charted percent or fraction entries, and PaO<sub>2</sub> values were standardized to mmHg. Each PaO<sub>2</sub> was paired to the nearest FiO<sub>2</sub> within a  $\pm 2$  hour window; pairs with implausible or missing FiO<sub>2</sub> were discarded. For each stay, the PF count, median, and minimum were aggregated over the 0–24 hour window.

Construction of the feature table was supported by an indexed cohort window. ICU stays with a documented out-time and length of stay of at least 24 hours were included. A 24 hour window was anchored at the ICU `intime`; all day-1 features were restricted to timestamps in `[intime, intime + 24h)`. Item-to-feature mappings were defined explicitly for core section-7 labs and label-driven for an extended panel. After normalization, per-analyte statistics were computed using exact medians (`PERCENTILE_DISC(0.5)`), minima, maxima, and means of indicator flags to form QC fractions. First and last values were obtained via time-ordered selection, and changes (*delta*) were computed only when at least two observations were available. Final curated panels were left-joined by `stay_id` to assemble a single, primary-keyed table that retains `hadm_id` and `subject_id` for relational linkage.

Naming conventions followed `snake_case` with descriptive bases and standardized suffixes (Table B.8). Vital signs used canonical 24 hour medians as default signals (additional suffix features were included where available). Therapy/exposure and Charlson variables were encoded as binary indicators with cohort-level prevalences documented. Where severity scores (e.g., APACHE III; SOFA domain totals and overall) were present, values were stored as point totals consistent with their original scales.

Table B.1: Endpoint and timing variables present in file.

| Column                              | Meaning (concise)                                           | Unit |
|-------------------------------------|-------------------------------------------------------------|------|
| Death                               | In-hospital death indicator ( <b>1=death</b> , 0=censored). | —    |
| Time to death                       | Hours from ICU admit to death / last alive (censored).      | h    |
| Time to AKI onset                   | Hours to AKI onset (KDIGO).                                 | h    |
| Time to sepsis onset                | Hours to first Sepsis–3 flag.                               | h    |
| Time to mechanical ventilator start | Hours to first invasive ventilation.                        | h    |
| Time to vasopressor start           | Hours to first vasopressor.                                 | h    |
| Age                                 | Age at ICU admission.                                       | y    |
| Gender                              | Biological sex ( <b>1=male</b> , 0=female).                 | —    |
| Admission weight                    | Documented admission weight (harmonized).                   | kg   |
| AKI trigger                         | Trigger type ( <b>1=creatinine</b> , 0=urine output).       | —    |

Table B.2: Therapy/exposure flags (definitions mirror Table 5.2).

| Column                             | Meaning (concise)                                       | Unit |
|------------------------------------|---------------------------------------------------------|------|
| Mechanical ventilation 24 h flag   | Invasive mechanical ventilation within T0–T24 (1/0).    | —    |
| Vasopressor 24 h flag              | Any vasopressor within T0–T24 (1/0).                    | —    |
| RRT 24 h flag                      | Renal replacement therapy within T0–T24 (1/0).          | —    |
| Baseline creatinine estimated flag | Baseline creatinine was estimated (vs. observed) (1/0). | —    |

Table B.3: Charlson comorbidity flags (1/0) and prevalence in cohort.

| Column                               | Meaning (concise)                       | Prevalence              |
|--------------------------------------|-----------------------------------------|-------------------------|
| Myocardial infarction flag           | History of myocardial infarction.       | 12.8%<br>(1,282/10,036) |
| Congestive heart failure flag        | Congestive heart failure.               | 11.8%<br>(1,187/10,036) |
| Peripheral vascular disease flag     | Peripheral vascular disease.            | 7.9%<br>(789/10,036)    |
| Cerebrovascular disease flag         | Stroke or TIA.                          | 9.7%<br>(975/10,036)    |
| Dementia flag                        | Dementia.                               | 4.1%<br>(412/10,036)    |
| Chronic pulmonary disease flag       | Chronic pulmonary disease (e.g., COPD). | 14.3%<br>(1,440/10,036) |
| Rheumatic / connective tissue flag   | Connective tissue disease.              | 2.1%<br>(214/10,036)    |
| Peptic ulcer disease flag            | Peptic ulcer disease.                   | 2.6%<br>(256/10,036)    |
| Liver disease (mild) flag            | Mild liver disease.                     | 10.4%<br>(1,043/10,036) |
| Diabetes (no complications) flag     | Diabetes without chronic complications. | 18.6%<br>(1,866/10,036) |
| Diabetes with complications flag     | Diabetes with end-organ complications.  | 9.3%<br>(935/10,036)    |
| Renal disease flag                   | Moderate–severe chronic kidney disease. | 14.2%<br>(1,421/10,036) |
| Hemiplegia/paraplegia flag           | Hemiplegia or paraplegia.               | 3.5%<br>(349/10,036)    |
| Cancer (no metastasis) flag          | Solid tumor within 5 y, no metastasis.  | 8.9%<br>(894/10,036)    |
| Metastatic cancer flag               | Metastatic solid tumor.                 | 3.9%<br>(392/10,036)    |
| HIV/AIDS flag                        | HIV/AIDS.                               | 0.4%<br>(36/10,036)     |
| Liver disease (moderate/severe) flag | Moderate or severe liver disease.       | 6.5%<br>(656/10,036)    |
| Charlson total                       | Sum of all Charlson flags (points).     | points                  |

Table B.4: Vital-sign base names; day-1 statistics derived via suffixes where applicable.

| Base name                | Meaning (concise)                                                           | Unit        |
|--------------------------|-----------------------------------------------------------------------------|-------------|
| Heart rate               | Day-1 aggregates (median, IQR, range, first/last, delta, slope, AUC, $n$ ). | beats/min   |
| Systolic blood pressure  | Systolic arterial blood pressure (non-/invasive).                           | mmHg        |
| Diastolic blood pressure | Diastolic arterial blood pressure (non-/invasive).                          | mmHg        |
| Mean arterial pressure   | Computed or recorded MAP.                                                   | mmHg        |
| Respiratory rate         | Respiratory rate.                                                           | breaths/min |
| Oxygen saturation        | Peripheral oxygen saturation (pulse oximetry).                              | %           |
| ShockIndex               | Derived ratio HR/SBP; higher reflects relative hypotension.                 | —           |

Table B.5: Fluid balance and urine output features.

| Column                         | Meaning (concise)                                     | Unit  |
|--------------------------------|-------------------------------------------------------|-------|
| Fluid balance 24 h (mL)        | Net inputs minus outputs in first 24 h (all sources). | mL    |
| Fluid balance 24 h per kg      | Weight-indexed net fluid balance (day 1).             | mL/kg |
| Urine output total 24 h        | Total urine in first 24 h.                            | mL    |
| Urine output total 24 h per kg | Weight-indexed urine output (day 1).                  | mL/kg |

Table B.6: Laboratory and gas-exchange base names (A–M).

| Base name                  | Meaning (concise)                       | Unit   |
|----------------------------|-----------------------------------------|--------|
| Albumin                    | Serum albumin.                          | g/dL   |
| Alkaline phosphatase       | Alkaline phosphatase.                   | U/L    |
| Aspartate aminotransferase | AST.                                    | U/L    |
| Anion gap                  | Serum anion gap.                        | mmol/L |
| Bicarbonate                | Serum bicarbonate ( $\text{HCO}_3^-$ ). | mmol/L |
| Blood urea nitrogen        | BUN.                                    | mg/dL  |
| Calcium (total)            | Total serum calcium.                    | mg/dL  |
| Chloride                   | Serum chloride.                         | mmol/L |
| Creatinine                 | Serum creatinine.                       | mg/dL  |
| Direct bilirubin           | Direct bilirubin.                       | mg/dL  |
| Glucose                    | Plasma glucose.                         | mg/dL  |
| Hemoglobin                 | Hemoglobin concentration.               | g/dL   |
| INR                        | International normalized ratio.         | ratio  |
| Lactate                    | Serum lactate.                          | mmol/L |
| LDH                        | Lactate dehydrogenase.                  | U/L    |

Table B.7: Laboratory and gas-exchange base names (N–Z).

| Base name              | Meaning (concise)                            | Unit            |
|------------------------|----------------------------------------------|-----------------|
| Magnesium              | Serum magnesium.                             | mg/dL           |
| $\text{PaCO}_2$        | Arterial carbon dioxide (ABG).               | mmHg            |
| $\text{PaO}_2$         | Arterial oxygen (ABG).                       | mmHg            |
| pH (arterial)          | Arterial pH (ABG).                           | pH units        |
| PF ratio               | $\text{PaO}_2/\text{FiO}_2$ ratio (derived). | mmHg            |
| Phosphate              | Serum phosphate.                             | mg/dL           |
| Platelet count         | Platelet count.                              | $10^9/\text{L}$ |
| Potassium              | Serum potassium.                             | mmol/L          |
| RDW                    | Red cell distribution width.                 | %               |
| Sodium                 | Serum sodium.                                | mmol/L          |
| Total bilirubin        | Total bilirubin.                             | mg/dL           |
| Troponin               | Cardiac troponin.                            | ng/mL           |
| White blood cell count | WBC.                                         | $10^9/\text{L}$ |

Table B.8: Suffix legend for time-series features.

| Suffix        | Definition (first 24 h)                                  | Unit   |
|---------------|----------------------------------------------------------|--------|
| <i>first</i>  | First measurement in the window.                         | base   |
| <i>last</i>   | Last measurement in the window.                          | base   |
| <i>median</i> | Median of all readings over 24 h.                        | base   |
| <i>iqr</i>    | Inter-quartile range (Q3–Q1).                            | base   |
| <i>rng</i>    | Range = max–min.                                         | base   |
| <i>delta</i>  | Change = last–first (requires $n \geq 2$ ).              | base   |
| <i>auc</i>    | Trapezoidal area-under-curve (0–24 h).                   | base·h |
| <i>slope</i>  | Linear regression slope per hour (requires $n \geq 2$ ). | base/h |
| <i>n</i>      | Number of discrete observations.                         | count  |

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